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Satellite Data Analysis for Heat Island Detection and Mitigation

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ABSTRACT

This thesis investigates the detection of heat islands and its insights using satellite data by creating an easy-to-use tool for local administrations and policymakers to help the environment.

The introduction chapter discusses the origins and significance of the heat island concept, emphasizing the need for effective detection and mitigation. It defines heat islands, examines their causes and effects on human health and the environment, and highlights the importance of sustainable urban development. The chapter reviews measurement and analysis techniques like satellite remote sensing and ground-based observations, stressing the need for integrated tools for land cover and air quality assessment. It provides an overview of the latest research and technologies in heat island detection, detailing the technical aspects of satellite data, particularly the Landsat and MODIS programs, and the trade-offs in using different satellite missions for land surface temperature and land cover analysis.

Chapter four outlines methods for analyzing land surface temperature data, detecting heat islands, and integrating land cover insights, including the use of OpenStreetMap for boundary retrieval and generating GPT-based recommendations for urban planners. The following chapter discusses the development of the Heat Island Insights Tool, detailing its architecture, backend services with Java Spring Boot, frontend interface with React, and data handling. It also covers the deployment strategy using AWS Cloud, Terraform, Ansible, and Docker. The conclusion highlights the importance of integrated tools for urban heat island mitigation, the need for advancements in satellite data, and the benefits of using Google Earth Engine. Future directions include enhancing real-time data integration, air quality insights, and specialized NLP models.

The contributions of this research lie in integrating advanced geospatial analysis, remote sensing data, and AI-driven insights to provide a robust framework for urban heat island detection and mitigation. This methodology supports effective strategies for creating sustainable urban environments.

This work is the result of my own activity. I have neither given nor received unauthorized assistance on this work.

David Simonel-Olimpiu



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Chapter 1

Introduction

Considering the evolution of society, there is an increasing demand for comprehensive information about the surrounding world. From the early days of civilization, humans have been inherently curious about the weather, which eventually spurred space exploration through artificial satellites. This milestone in human achievement marked the beginning of space exploration. However, despite the allure of the vast expanse beyond our planet, most people have remained more intrigued by understanding the intricacies of the Earth's surface. This focus on our immediate environment has paved the way for developing the Digital Twin concept for cities, a revolutionary approach that promises to enhance our understanding of urban environments.

The concept of a Digital Twin, which refers to a virtual replica of a physical entity, was first introduced in 2002 by Dr. Michael Grieves during a presentation at the University of Michigan[1]. Initially conceived within a Product Lifecycle Management center, this concept has evolved significantly. Over the past two decades, the idea of a Digital Twin City has matured, garnering attention from large corporations, administrative institutions, and urban planners. This digital representation of a city is invaluable for understanding the impact of construction and urban development on the environment. It also aids individuals in making informed decisions about where to live by considering various factors such as educational facilities, healthcare institutions, employment opportunities, and recreational spaces.

One of the critical factors influencing the livability and comfort of a city is its mean temperature. This factor affects indoor temperatures and can vary depending on whether a hose faces the sun. Moreover, the proximity to green spaces like parks, population density, and industrial zones play significant roles in determining air quality and overall heat levels in urban areas. These considerations have led to the emergence of the concept of Urban Heat Islands (UHIs) and the broader notion of Heat Islands, which describe areas that experience significantly higher temperatures than their rural surroundings due to human activities and modifications to the

natural environment.

Through my extensive research and studies, I discovered that analyzing heat maps can reveal not only the typical urban heat islands within cities but also unexpected hot spots such as landfills and construction sites. By integrating a Heat Island Detection Algorithm with a Land Coverage Algorithm and an Air Quality Algorithm, all utilizing satellite data, these heat islands can be effectively identified, labeled, and reported to relevant administrative entities. This multi-faceted approach provides a comprehensive understanding of the factors contributing to urban heat and facilitates the implementation of mitigation strategies.

This thesis presents a detailed exploration of the algorithms and free satellite data available as of the end of 2023. It outlines the development of a sophisticated software application designed to merge this diverse information into a unified platform. This platform enables the visualization of heat islands and their potential causes, providing valuable insights for urban planners and policymakers. The research primarily focuses on the software application that is flexible, allowing it to automatically analyze any administrative area based on its name, time frame, and location type. In addition to the primary focus on heat island detection, this paper delves into the broader implications of urban heat management. It examines how integrating various datasets can lead to more effective urban planning and environmental conservation strategies. The discussion also covers the potential benefits of such a comprehensive tool in mitigating the adverse effects of urban heat islands, improving air quality, and enhancing city dwellers' overall quality of life. The significance of this research lies in its potential to bridge the gap between technological advancements and practical applications in urban management. This study aims to provide a robust framework for detecting and addressing heat islands by leveraging cutting-edge satellite data and sophisticated algorithms. This, in turn, can lead to more sustainable and resilient urban environments, better equipped to handle the challenges of climate change and rapid urbanization.

This thesis aims to contribute to the growing body of knowledge on urban heat islands and their impact on urban environments. It seeks to provide a practical tool that can assist cities worldwide in managing heat more effectively, ensuring a better and more sustainable future for urban populations.

1.1 Usage of AI generative tools

During the preparation of this work, the author used ChatGPT to structure the information in chapters 2 and 3 (the theoretical chapters). After using this tool, the author reviewed, edited, and elaborated on the generated content and assumes full responsibility for the content of the work.

Chapter 2

State of the Art

2.1 Problem Statement

2.1.1 Current Challenges

Despite significant advancements in understanding and mitigating urban heat islands, there is a critical need for an integrated analytical tool that combines heat island detection, land cover analysis, and air quality assessment. Current methodologies often rely on disparate systems and fragmented data sources, making comprehensive environmental analysis cumbersome and inefficient[2].

Current methods typically focus on single aspects such as temperature or land cover without integrating other crucial environmental factors. For instance, systems that detect heat islands using thermal imaging often fail to correlate these findings with land use patterns or air quality data. This disjointed approach results in an incomplete understanding of the environmental dynamics at play, hindering the development of effective mitigation strategies[3].

Moreover, the lack of integration leads to inefficiencies and potential inaccuracies in environmental assessments. For example, remote sensing data from satellites such as Landsat and MODIS provide valuable information on land surface temperatures and land cover changes, but without integration with air quality measurements, the analysis remains partial and limited in scope[2]. Advanced computational techniques, including machine learning and artificial intelligence, are often underutilized in analyzing these large datasets, which could otherwise reveal patterns and trends not apparent through traditional methods[4].

The fragmented nature of current approaches limits the ability to create predictive models necessary for forecasting future environmental conditions. This gap in comprehensive analysis tools impedes urban planners and policymakers from effectively anticipating and mitigating the impacts of various interventions, such as introducing green spaces or changes in land use policies[3].

Heat islands contribute to increased temperatures and exacerbate the formation of ground-level ozone and other pollutants, significantly impacting air quality. Poor air quality is a critical factor contributing to various health issues, including respiratory and cardiovascular diseases. Additionally, higher temperatures can lead to heat stress and heat-related illnesses, which are particularly dangerous for vulnerable populations such as the elderly and children[2, 4].

In summary, the problem lies in the absence of a unified tool that can seamlessly integrate heat island detection, land cover analysis, and air quality assessment to provide a complete and accurate understanding of urban environmental challenges.

2.1.2 Heat Island Causes on Humans and Environment

Heat islands, particularly urban heat islands (UHIs), result in localized regions of elevated temperatures in urban areas compared to their rural surroundings. These temperature differences arise from the extensive use of materials like concrete, asphalt, and buildings that absorb and retain heat. Human activities, such as industrial processes and transportation, further contribute to the heat island effect[5, 2].

- **Impact on Human Health**

The impact of heat islands on human health is significant. Increased temperatures can exacerbate heat-related illnesses like heat stroke, dehydration, and cardiovascular problems. Vulnerable populations, including the elderly, children, and those with pre-existing health conditions, are particularly at risk. Prolonged exposure to high temperatures can also lead to increased mortality rates during heatwaves[6].

Higher temperatures in urban areas can also negatively affect mental health. Studies have shown that extreme heat can increase stress levels and decrease cognitive function, impacting overall well-being and productivity[6].

- **Environmental Impacts**

The environmental impacts of heat islands are wide-ranging. Elevated temperatures can lead to increased energy consumption as the demand for air conditioning rises. This results in higher greenhouse gas emissions and contributes to the urban heat island effect by releasing waste heat into the atmosphere[7].

Heat islands also affect local ecosystems. Higher temperatures can alter the growth patterns of urban vegetation, reduce biodiversity, and impact water quality by increasing the temperature of water bodies[7]. Additionally, the formation of urban smog is accelerated by higher temperatures, further degrading air quality[6].

- **Benefits of Cold Spots**

In contrast, cold spots provide several benefits. Areas with abundant vegetation and water bodies, often called cold spots, can significantly reduce local temperatures. Vegetation cools the air through the process of evapotranspiration, where water absorbed by plant roots is released into the atmosphere, lowering temperatures. Trees and plants also provide shade, further reducing heat absorption by urban surfaces[8].

Water bodies, such as lakes, rivers, and ponds, help moderate temperatures as they absorb heat more slowly than land and release it gradually. This thermal regulation helps maintain a cooler microclimate in surrounding areas. Cold spots also improve air quality by filtering pollutants and increasing oxygen levels, contributing to better human health and enhanced biodiversity[8, 7].

2.2 Introduction to Heat Islands

The term "Heat Islands" refers to areas that exhibit higher temperatures than their surrounding environments. This phenomenon can be observed in various contexts, including urban, suburban, and even rural locales, leading to a broader interpretation beyond the commonly discussed Urban Heat Island (UHI) effect. At its core, the heat island effect results from the differential heating of surfaces, where certain materials or land covers retain and radiate heat more than others[9].

The heat island effect basics are grounded in solar radiation absorption and thermal emission principles. Surfaces like concrete, asphalt, and buildings absorb more solar energy during the day and release it at night, leading to elevated temperatures than nearby less-developed or vegetated areas. This thermal behavior contributes to the temperature discrepancies that define heat islands[10].

Historically, the concept of heat islands has been explored since the 1800s, but it has gained significant attention in modern times due to urban expansion and its associated environmental impacts. Heat islands are not exclusively an urban phenomenon; they also occur in other modified landscapes, such as industrial zones or areas with extensive non-natural surfaces[3].

Recognizing and understanding heat islands are essential for addressing their environmental, health, and ecological challenges. This includes influencing local weather patterns, contributing to pollution levels, and impacting human health and comfort.

2.2.1 Types of Heat Islands

Heat Islands are classified mainly based on their location and the materials involved. The most recognized types are Urban Heat Islands (UHI) and Surface Heat

Islands (SHI), but distinctions also extend to peri-urban, rural, and water body heat islands, each with unique characteristics and driving factors.

- **Urban Heat Islands**

Urban Heat Islands (UHI) are a prominent feature of modern urban environments, where localized warming effects lead to higher temperatures in urban areas compared to their rural surroundings. This phenomenon is primarily due to the dense concentration of buildings, roads, and other infrastructure that absorb and retain solar energy. Materials like concrete and asphalt, commonly used in urban construction, have high thermal mass and low albedo, contributing significantly to the UHI effect[9]. Additionally, the anthropogenic heat generated from vehicles, industrial activities, and energy consumption for heating and cooling buildings exacerbates the temperature rise in urban centers[5].

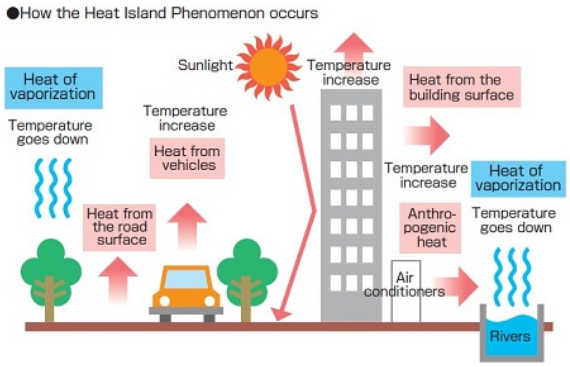


Figure 2.1: Urban Heat Island Temperature Reflection[11]

The impact of UHI on cities is multifaceted, affecting energy consumption, air quality, and public health. Urban areas with pronounced UHI effects often experience increased demand for air conditioning, leading to higher energy usage and associated greenhouse gas emissions. The elevated temperatures can also accelerate the formation of urban smog, exacerbating pollution levels and posing health risks to the urban population. Moreover, heat stress and heat-related illnesses become more prevalent in these overheated urban environments, highlighting the need for effective UHI mitigation strategies[6].

- **Surface Heat Islands**

Surface Heat Islands (SHI) arise from the differential heating of ground surfaces, where materials like asphalt, dark roofs, and concrete exhibit higher temperatures due to their physical properties. SHIs are particularly significant in both urban and rural settings, where their impact can vary based on the material composition, color, and usage of the land surface[12]. The phenomenon is closely related to the concept

of albedo, the measure of a surface's ability to reflect solar energy. Surfaces with low albedo absorb more heat, contributing to localized temperature increases and influencing the microclimatic conditions[8].

- **Peri-urban Heat Islands**

Peri-urban Heat Islands occur at the interface between urban and rural areas, where land-use changes and urban sprawl lead to increased surface temperatures. These areas, characterized by a mix of agricultural land, fragmented natural habitats, and expanding urban development, are critical zones for studying the extension of UHI effects into the countryside[4].

- **Rural Heat Islands**

Rural Heat Islands (RHI) are less commonly discussed but can significantly affect local climates and ecosystems. These phenomena occur in rural areas where natural land cover is replaced by surfaces like asphalt, concrete, or bare soil, leading to increased absorption of solar radiation and higher surface temperatures. Agricultural activities, such as large-scale cropping or livestock farming, can also contribute to RHI by altering land cover and soil properties, thereby affecting the thermal characteristics of the area[13].

- **Water Body Heat Islands**

Water Body Heat Islands (WBHI) manifest as increased water temperatures in lakes, rivers, and reservoirs near urbanized or industrial areas. These temperature elevations are often due to factors such as thermal pollution from industrial discharges, urban runoff heating during hot weather, and removing shading vegetation along water bodies[7].

2.2.2 Heat Island Causes

- **Causes of Heat Islands**

The primary causes of heat islands include the alteration of land surfaces and the accumulation of anthropogenic heat. Urbanization replaces natural landscapes with impervious surfaces like concrete and asphalt, with high thermal masses and low albedo, leading to increased heat absorption and retention[9, 7]. Additionally, activities such as transportation, industrial operations, and energy consumption release significant amounts of waste heat into the environment[5, 2].

The construction of buildings and roads also contributes to the heat island effect. These structures absorb and store heat during the day and release it at night, causing

elevated temperatures throughout the urban area. The lack of vegetation and green spaces further exacerbates this effect by reducing the natural cooling provided by plants and trees[8].

- **Causes of Cold Spots**

Cold spots, on the other hand, are primarily caused by the presence of vegetation and water bodies. Plants and trees provide shade and cool the air through the process of evapotranspiration, where water absorbed by roots is released into the atmosphere, reducing temperatures. Large green spaces, such as parks and urban forests, act as natural coolers, mitigating the heat island effect[8, 7].

Water bodies also play a crucial role in creating cold spots. They absorb heat more slowly than land and release it gradually, helping to moderate temperatures. The presence of water bodies can significantly influence local microclimates, providing cooler and more comfortable environments for urban residents[7].

2.3 Case Studies and Heat Island Analysis

2.3.1 Case Studies

This section presents case studies demonstrating the effectiveness of UHI mitigation and adaptation strategies.

- **Case Study 1: Green Roof Implementation in New York City**

A study in New York City showed that the installation of green roofs significantly reduced surface temperature. Thermal imaging and environmental sensor data indicated a temperature differential of up to 3°C during peak summer[2], as calculated by equation 2.1.

$$\Delta T = T_{\text{conventional}} - T_{\text{green}} \tag{2.1}$$

- **Case Study 2: Urban Park Cooling Effect in Chicago**

Research in a large urban park in Chicago demonstrated its cooling effect on surrounding neighborhoods. Remote sensing data and GIS analysis showed a cooling benefit extending up to 500 meters from the park boundary, with a maximum temperature difference of 4°C compared to areas without green space[14], modeled using equation 2.2.

$$T_d(x) = T_0 \exp\left(-\frac{x}{L}\right) \tag{2.2}$$

- **Case Study 3: Reflective Surfaces in Los Angeles**

A citywide initiative in Los Angeles to replace traditional pavements and roofing materials with high-albedo alternatives resulted in measurable reductions in UHI intensity. The study found an average surface temperature decline of 2°C across treated areas[12], calculated through 2.3.

$$T_{\text{new}} = T_{\text{old}} - \Delta T_{\text{reflective}} \tag{2.3}$$

The measurement and analysis of heat islands necessitate a multi-faceted approach, integrating remote sensing, ground-based observations, and computational techniques. This section elucidates the methodologies employed in capturing and analyzing the nuanced dynamics of urban heat islands (UHI), with a particular emphasis on the Landsat mission for its pivotal role in land surface temperature (LST) and land use data collection[3].

2.3.2 Measurement and Analysis for Heat Islands

- **Satellite Remote Sensing**

Landsat 8's thermal infrared (TIR) sensor data are instrumental in deriving accurate LST, a key parameter for UHI analysis. This process encompasses atmospheric correction steps to convert digital numbers (DN) into spectral radiance and then to brightness temperature, providing a base for heat island intensity calculations[15]. The radiative transfer equation used in this context reflects the interaction between land surface and atmospheric properties.

$$L_\lambda = \frac{M_L \times DN + A_L}{\cos\theta} \tag{2.4}$$

In the equation 2.4, L_λ signifies the spectral radiance, M_L denotes the band-specific multiplicative rescaling factor, DN represents the digital number, A_L is the additive rescaling factor, and θ is the satellite zenith angle. This technical process is crucial for transforming raw satellite data into a meaningful metric that reflects the actual temperature of the Earth's surface[7].

- **Ground-Based Observations**

Ground-based temperature measurements are essential for validating and calibrating the satellite-derived LST data. Weather stations and mobile sensors scattered across urban and rural settings provide precise temperature readings that help fine-tune the satellite data. These localized measurements are critical for ensuring the accuracy and reliability of the LST values used in UHI studies[9].

- **Data Analysis and Computational Techniques**

Analyzing the heat island intensity necessitates a comprehensive comparison between the LST of urban and surrounding rural areas. GIS-based techniques are employed to delineate urban boundaries and facilitate the extraction and comparison of temperature data[16].

$$UHI_{intensity} = T_{urban} - T_{rural} \tag{2.5}$$

The calculation 2.5 of UHI intensity serves as a quantitative measure of the heat island effect, informing urban climatology studies and mitigation strategies. Integrating LST data with urban morphology, land use, and meteorological information via advanced computational models, including machine learning algorithms, enhances the prediction and analysis of UHI phenomena[4, 8].

- **Multisource Data Integration**

Incorporating multisource data is paramount in constructing a holistic view of UHI. Landsat 8's data, while foundational, are augmented by inputs from other satellite missions like the Copernicus Sentinel series, which offers high-resolution imagery and varied observational capabilities, thus enriching the analysis of urban landscapes and their thermal characteristics[3, 15].

Socio-economic and meteorological data, when synthesized with satellite imagery, illuminate the intricate relationships between urban development, land use changes, and climatic conditions in the context of heat islands. This multidimensional dataset provides a robust foundation for exploring the underlying mechanisms of UHI and devising targeted intervention strategies[2].

- **Validation and Calibration**

Ensuring the accuracy of satellite-derived LST data is imperative for reliable UHI analysis. Calibration against ground-based measurements ensures the veracity of satellite readings, correcting for any atmospheric interference or sensor inaccuracies[3, 7].

Regular validation exercises, aligned with international standards and protocols, help maintain the integrity of the data used in UHI studies. Such practices are essential for ensuring the scientific rigor of the analysis, supporting the development of sound, evidence-based urban planning and policy-making strategies[17].

2.4 Mitigations and Adaptation Strategies

Addressing the adverse effects of heat islands requires a combination of mitigation and adaptation strategies. Mitigation focuses on reducing heat sources, while adaptation involves adjusting to the impacts of increased temperatures.

2.4.1 Mitigation Strategies

Increasing Urban Greenery

Increasing urban greenery through parks, green roofs, and street trees is one of the most effective mitigation strategies. Vegetation provides shade and cools the air through evapotranspiration, reducing the overall temperature in urban areas. Green roofs, which involve growing plants on rooftops, can significantly lower rooftop temperatures and improve building insulation[2, 14].

Cool Roofs and Reflective Pavements

Implementing cool roofs and reflective pavements with high albedo materials can reduce heat absorption. These materials reflect more sunlight and absorb less heat, lowering surface temperatures and mitigating the heat island effect. Cool roofs, in particular, can reduce indoor temperatures, decreasing the need for air conditioning and energy consumption[2].

Improving Building Energy Efficiency

Enhancing building energy efficiency is crucial for reducing waste heat. Using advanced insulation materials, energy-efficient HVAC systems, and smart building technologies can lower energy consumption and reduce the heat emitted by buildings. This approach mitigates the heat island effect and contributes to overall energy savings[5].

2.4.2 Adaptation Strategies

Urban Design and Planning

Adaptation strategies involve modifying urban designs to improve airflow and reduce heat accumulation. Designing cities with adequate green spaces, open areas, and water bodies can create cooler microclimates. Urban planning should also focus on creating compact and energy-efficient building layouts to minimize heat retention[6, 13].

Heat Action Plans

Developing heat action plans is essential for preparing cities for extreme heat events. These plans include measures such as establishing cooling centers, providing public health advisories, and implementing emergency response protocols. Raising public awareness about the risks of heat and promoting heat safety measures are also critical components of heat action plans[6].

Infrastructure Resilience

Enhancing infrastructure resilience to withstand extreme temperatures is vital. This involves using heat-resistant materials in construction, ensuring a reliable energy supply during heatwaves, and maintaining critical infrastructure like water and transport systems. Integrating climate considerations into urban planning and infrastructure development helps cities adapt to the changing climate[13].

2.5 Heat Island Detection. Land Cover and Air Quality Analysis

2.5.1 Introduction to Satellite Data

Satellite data plays a crucial role in environmental monitoring due to its ability to provide comprehensive and consistent coverage of the Earth's surface. Unlike ground-based observations, satellites can monitor large and remote areas, providing data essential for understanding global environmental changes[3]. This capability is particularly important for tracking phenomena such as deforestation, climate change, and urbanization[16].

Satellites equipped with various sensors can capture data in different spectral bands, allowing for the analysis of diverse environmental parameters. For instance, optical sensors can be used to monitor vegetation health, while thermal infrared sensors can detect land surface temperature, which is critical for studying urban heat islands (UHI)[2]. Integrating these data sources provides a more complete picture of environmental conditions and trends[15].

Furthermore, satellite data supports the development of predictive models and early warning systems, which are essential for disaster management and mitigation[17]. By providing timely and accurate information, satellite data helps policymakers and researchers make informed decisions to protect and manage natural resources effectively[5].

2.5.2 Algorithms for Heat Island Detection

The detection of heat islands is crucial for understanding and mitigating urban heat island (UHI) effects. The process involves using satellite images to calculate the land surface temperature (LST) and identifying areas with higher temperatures than their surroundings. This algorithm combines the Getis-Ord-Gi* statistic and the standard deviation of the Normalized Difference Vegetation Index (NDVI) to extract UHI areas from Landsat images [18]. Landsat images provide high-resolution thermal data necessary for this analysis.

The algorithm 1 begins by calculating the LST from the Landsat images. Next, the NDVI and its standard deviation are computed to incorporate vegetation cover information, which is inversely related to surface temperature. The Getis-Ord-Gi* statistic is then applied to the LST to identify hotspots. These hotspots are refined using the NDVI standard deviation to delineate heat island areas accurately. The final step involves converting these areas into vector polygons for further analysis and visualization.

Data: LandsatImages, AOI
Result: HeatIslandPolygons

Calculate Land Surface Temperature (LST) from Landsat images;
LST ← CalculateLST(LandsatImages);

Calculate NDVI and its standard deviation (NDVI STD);
NDVI ← CalculateNDVI(LandsatImages);
NDVI_STD ← CalculateNDVI_STD(NDVI);

Apply Getis-Ord-Gi* to LST to identify hotspots;
UHI_hotspots ← GetisOrdGi*(LST);

Combine UHI hotspots with NDVI STD to refine UHI areas;
HeatIslandAreas ← Combine(UHI_hotspots, NDVI_STD);

Convert UHI areas to vector polygons;
HeatIslandPolygons ← ConvertToPolygons(HeatIslandAreas);

Algorithm 1: Heat Island Detection Algorithm

This method leverages the high-resolution thermal bands from the Landsat series, specifically Landsat 8, which provides the Operational Land Imager (OLI) and Thermal Infrared Sensor (TIRS) data essential for LST and NDVI calculations. The Getis-Ord-Gi* statistic is a well-established method for spatial analysis, enabling the identification of statistically significant hotspots [18].

2.5.3 Algorithms for Land Cover Analysis

Land cover analysis is vital for understanding the distribution and changes in land use over time. MODIS (Moderate Resolution Imaging Spectroradiometer) data is commonly used for this purpose due to its high temporal resolution and global coverage. The algorithm 2 involves clipping the MODIS land cover data to the area of interest (AOI) and calculating the area and percentage of each land cover type within the AOI [19].

The first step is to define the AOI using geographic coordinates. The MODIS land cover data is then clipped to this AOI to focus the analysis. The area of each land cover type is calculated within the AOI, followed by the calculation of percentage coverage for each type. This provides a detailed understanding of the land cover distribution and its dynamics.

Data: MODISLandCoverImage, AOI
Result: LandCoverStatistics

Define AOI using geographic coordinates;
aoi ← DefineAOI(Coordinates);

Clip MODIS land cover data to AOI;
clippedLandCover ← ClipToAOI(MODISLandCoverImage, aoi);

Calculate the area of each land cover type within AOI;
landCoverAreas ← CalculateAreas(clippedLandCover);

Calculate percentage coverage of each land cover type;
landCoverPercentages ← CalculatePercentages(landCoverAreas, aoi);

Algorithm 2: Land Cover Analysis Algorithm

This algorithm uses MODIS data, specifically the MODIS Land Cover Type product (MCD12Q1), which provides yearly global land cover maps at 500m spatial resolution. The use of geographic coordinates to define the AOI ensures that the analysis is spatially accurate and relevant to the specific area under study [19].

2.5.4 Algorithms for Air Quality Analysis

Air quality analysis is essential for monitoring pollution levels and assessing their impact on public health and the environment. This algorithm integrates satellite data with ground-based measurements to calculate an air quality index (AQI). Satellite data, such as from the MODIS or Sentinel-5P missions, provide wide coverage and are complemented by ground-based data from monitoring stations to enhance accuracy [20].

The algorithm 3 begins by applying thresholds to the satellite data to identify polluted areas. These areas are then integrated with ground-based measurements to create a comprehensive dataset. The final step involves calculating the AQI for the AOI, providing a quantitative measure of air quality.

Data: SatelliteData, GroundData
Result: AirQualityIndex

Apply thresholds to satellite data to identify polluted areas;
pollutedAreas ← ApplyThreshold(SatelliteData, threshold);

Integrate satellite data with ground-based measurements;
combinedData ← IntegrateData(pollutedAreas, GroundData);

Calculate air quality index for AOI;
airQualityIndex ← CalculateAirQualityIndex(combinedData, AOI);

Algorithm 3: Air Quality Analysis Algorithm

This approach utilizes data from satellite missions like MODIS, which provides aerosol optical depth (AOD) measurements, and Sentinel-5P, which offers detailed information on atmospheric gases. Ground-based data from air quality monitoring stations are used to calibrate and validate the satellite observations, ensuring the accuracy of the AQI calculation [20].

2.5.5 Existing Tools

Currently, several tools are available for analyzing heat islands, land cover changes, and air quality, but none integrate all these aspects into a cohesive system. For example, Google Earth Engine (GEE) provides a powerful platform for geospatial analysis but is not tailored specifically for comprehensive heat island management [21]. Other tools, such as MODIS and Sentinel-5P, focus on air quality monitoring, while urban planning tools often lack integration with environmental data [22, 23].

The GEE platform, with its ability to handle large-scale data and perform complex analyses, has been utilized for continuous land change monitoring and environmental studies. However, it requires significant expertise to set up and operate effectively for specific applications like urban heat island (UHI) analysis [12]. Similarly, tools like ArcGIS Urban Heat Island Effect Dashboard provide detailed visualizations and interactive maps but do not offer comprehensive integration with air quality and land cover data [8].

The Heat Island Analysis Tool aims to fill this gap by integrating heat island detection, land cover analysis, and air quality assessment into one platform. Enhanced by a large language model (LLM), it provides actionable recommendations,

ensuring a comprehensive and efficient analysis. This novel approach significantly improves existing methodologies by combining these diverse data sources into a single, user-friendly interface [4, 10, 24].

Additionally, Sentinel-3 data has been employed for surface heat island detection using image segmentation techniques, but these methods are typically isolated and lack integration with other critical environmental parameters such as air quality [13]. Other innovative approaches include autonomous drones for air pollution monitoring, which offer real-time data but are not integrated with UHI and land cover analysis [11].

2.6 Relevance of a Heat Island Insights Tool

The proposed integrated tool is highly relevant in today's rapid urbanization and climate change context. By providing detailed and accurate environmental data, this tool can aid local administrations in developing strategies to mitigate the effects of heat islands. Integrating an LLM enables the analysis of complex datasets and the generation of tailored recommendations for urban planners and policymakers.

The tool's ability to combine heat island detection with land cover and air quality analysis offers a comprehensive approach to environmental management. This holistic view is essential for addressing the multifaceted impacts of urbanization and climate change, ensuring sustainable and resilient urban development [9, 2]. Moreover, the tool's real-time data and predictive analytics capabilities allow for proactive decision-making, helping cities to better prepare for and mitigate the effects of extreme heat events. By providing actionable insights and recommendations, the tool supports informed policy-making and efficient resource allocation, ultimately enhancing the livability and sustainability of urban areas [1, 6].

The existing tools, while powerful in their domains, often require specialized knowledge to operate and do not provide an integrated analysis of UHI, land cover changes, and air quality. For example, the continuous change detection and classification (CCDC) algorithm in GEE is excellent for monitoring land cover changes over time but lacks direct integration with real-time air quality data and specific UHI metrics [12]. Tools like the Sentinel-3 surface heat island detection framework focus on specific aspects without the broader environmental context [13].

By bridging these gaps, the proposed tool not only streamlines the analytical process but also democratizes access to advanced environmental insights, making it accessible for a wider range of users, including policymakers, urban planners, and researchers. This integration is crucial for formulating comprehensive strategies to combat urban heat islands and their associated challenges, ensuring more sustainable urban futures.

Chapter 3

Satellite Data

3.1 Structure and Function of Satellites

3.1.1 Main Components of Satellites (Payload, Bus, Antennas, Solar Panels)

Satellites are complex machines composed of several key components that work together to achieve their mission objectives. The four primary components of a satellite are the payload, the bus, the antennas, and the solar panels.

The payload is the part of the satellite that performs the mission's specific tasks, such as capturing images or collecting scientific data. In Earth observation satellites, the payload typically includes sensors and instruments designed to detect and measure various environmental parameters [25].

The bus, or platform, is the supporting structure that houses the payload and provides the necessary infrastructure for its operation. This includes power supply, thermal control, communication, and propulsion systems. The bus ensures the payload can function correctly and transmit the collected data back to Earth [26].

Antennas are crucial for communication between the satellite and ground stations. They transmit the data collected by the payload back to Earth and receive commands from mission control. High-gain antennas are used for long-distance communication, while low-gain antennas are used for short-distance communication and initial satellite tracking after launch [27].

Solar panels are the primary source of power for satellites. They convert sunlight into electrical energy, which is used to power the satellite's instruments, onboard computers, and communication systems. Solar panels are typically mounted on the satellite's exterior and are designed to track the Sun to maximize energy absorption [28].

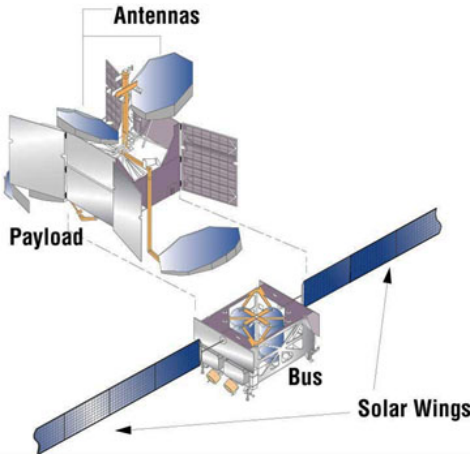


Figure 3.1: Diagram of satellite structure main components.

3.1.2 Types of Sensors and Instruments (Optical, Thermal Infrared)

The sensors and instruments onboard satellites are critical for capturing the data needed for various applications. The main types of Earth observation sensors include optical and thermal infrared sensors.

Optical sensors capture visible and near-infrared light reflected from the Earth's surface. These sensors are widely used for monitoring vegetation, land use, and water bodies. They provide high-resolution images that are essential for detailed mapping and analysis [29].

Thermal infrared sensors measure the emitted infrared radiation from the Earth's surface, which is related to temperature. These sensors are crucial for monitoring land surface temperature (LST), which is important for studying urban heat islands and other thermal phenomena [3].

3.1.3 Data Transmission and Processing Mechanisms

Once the data is captured by the sensors, it needs to be transmitted back to Earth for processing and analysis. This involves several steps and mechanisms to ensure the data's integrity and accuracy.

The data transmission process typically begins with the onboard storage of the collected data, which is then transmitted to ground stations via radio waves. These ground stations receive and relay the data transmitted to data processing centers [27].

At the data processing centers, the raw data undergoes various levels of processing to correct for sensor errors, atmospheric interference, and other factors. This processing transforms the raw data into usable products, such as calibrated images and derived environmental parameters, which can be used for analysis and decision-making [28].

The continuous improvements in data transmission and processing technologies have significantly enhanced the quality and accessibility of satellite data, making it a vital resource for environmental monitoring and research [30].

3.2 Landsat Program

3.2.1 Overview and Historical Background

The Landsat program, initiated by NASA in the early 1970s, is one of the most significant and enduring Earth observation programs. The primary objective of the Landsat program is to provide continuous, high-quality, moderate-resolution data of the Earth's surface to monitor and manage natural resources, track land use changes, and support scientific research [3]. The Earth Resources Technology Satellite (ERTS) program was originally named and later renamed Landsat in 1975 [31].

Landsat-1, launched in 1972, was the first satellite in the series, equipped with a multispectral scanner system (MSS) that provided unprecedented views of the Earth's surface. This satellite set the foundation for future missions, demonstrating the value of satellite remote sensing for environmental monitoring and management [30].

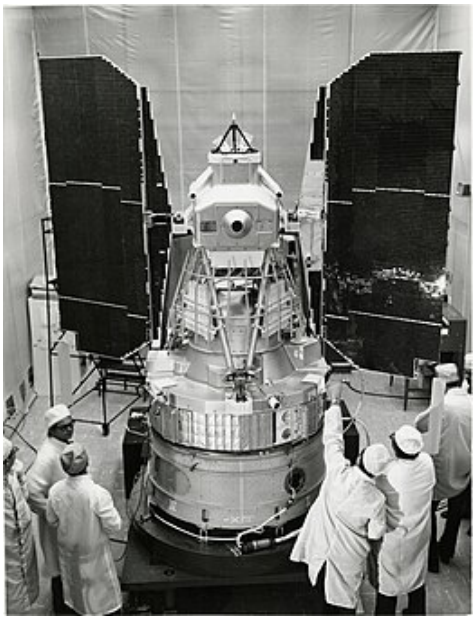


Figure 3.2: Landsat-1 satellite.

3.2.2 Evolution of Landsat Satellites (Key Improvements)

Over the decades, the Landsat program has seen significant advancements and improvements in satellite technology and sensor capabilities. Each successive Landsat

satellite has built upon the successes and lessons learned from its predecessors, incorporating new technologies and addressing emerging needs.

Landsat-2 and Landsat-3, launched in 1975 and 1978, respectively, continued the mission of Landsat-1 with similar capabilities but improved reliability and coverage. These satellites ensured the continuity of data collection and provided valuable insights into long-term environmental changes [31].

The introduction of the Thematic Mapper (TM) sensor on Landsat-4 and Landsat-5, launched in 1982 and 1984, respectively, marked a significant leap in sensor technology. The TM sensor offered higher spatial resolution and additional spectral bands, enhancing the ability to monitor various environmental parameters with greater precision [3].

Landsat-7, launched in 1999, introduced the Enhanced Thematic Mapper Plus (ETM+), which included a panchromatic band with a 15-meter resolution and a thermal infrared channel. This satellite improved image quality and enhanced the program's capacity for detailed environmental monitoring [30].

Landsat-8, launched in 2013, brought further advancements with the Operational Land Imager (OLI) and Thermal Infrared Sensor (TIRS). These sensors offer improved radiometric resolution, additional spectral bands, and enhanced data quality, supporting various applications from agriculture to urban planning [3].

3.2.3 Technical Specifications of Landsat Satellites

The technical specifications of Landsat satellites have evolved significantly since the launch of Landsat-1. Key improvements in spatial resolution, spectral bands, and radiometric accuracy have been made, enhancing the program's ability to capture detailed and accurate environmental data.

Landsat-1 to Landsat-3:

- **Sensors:** Multispectral Scanner System (MSS)
- **Spatial Resolution:** 80 meters
- **Spectral Bands:** 4 bands (Green, Red, Near-Infrared, Far-Infrared)
- **Temporal Resolution:** 18 days [31]

Landsat-4 and Landsat-5:

- **Sensors:** Thematic Mapper (TM), Multispectral Scanner System (MSS)
- **Spatial Resolution:** 30 meters (TM), 80 meters (MSS)
- **Spectral Bands:** 7 bands (Visible, Near-Infrared, Shortwave Infrared, Thermal Infrared)

- **Temporal Resolution:** 16 days [3]

Landsat-7:

- **Sensors:** Enhanced Thematic Mapper Plus (ETM+)
- **Spatial Resolution:** 15 meters (Panchromatic), 30 meters (Multispectral), 60 meters (Thermal)
- **Spectral Bands:** 8 bands (Visible, Near-Infrared, Shortwave Infrared, Thermal Infrared, Panchromatic)
- **Temporal Resolution:** 16 days [30]

Landsat-8:

- **Sensors:** Operational Land Imager (OLI), Thermal Infrared Sensor (TIRS)
- **Spatial Resolution:** 15 meters (Panchromatic), 30 meters (Multispectral), 100 meters (Thermal resampled to 30 meters)
- **Spectral Bands:** 11 bands (Visible, Near-Infrared, Shortwave Infrared, Thermal Infrared, Panchromatic, Cirrus)
- **Temporal Resolution:** 16 days [3]

The consistent improvements in the Landsat program have provided a valuable and continuous record of Earth's surface, supporting numerous applications in environmental monitoring, resource management, and scientific research.

3.3 MODIS Program

3.3.1 Overview and Historical Background

The MODIS (Moderate Resolution Imaging Spectroradiometer) program is a key component of NASA's Earth Observing System (EOS) that provides critical data for environmental and climate research. Launched aboard the Terra satellite in 1999 and the Aqua satellite in 2002, MODIS was designed to provide high temporal resolution data that complements the spatial resolution data of other satellite missions like Landsat [22].

The instruments capture data in 36 spectral bands, including visible, near-infrared, shortwave, and thermal infrared regions. This wide range of spectral capabilities enables comprehensive monitoring of Earth's atmosphere, oceans, and land surfaces [32].



Figure 3.3: MODIS satellite on the Terra platform.

3.3.2 Technical Specifications of MODIS Satellites

MODIS has advanced sensors that provide critical data for various applications. The key technical specifications are:

- **Spatial Resolution:** 250 meters (Bands 1-2), 500 meters (Bands 3-7), 1000 meters (Bands 8-36)
- **Spectral Bands:** 36 bands (Visible, Near-Infrared, Shortwave Infrared, Thermal Infrared)
- **Temporal Resolution:** 1-2 days (global coverage) [22]

A high temporal resolution with near-daily global coverage is provided, essential for monitoring dynamic changes in the environment. The spectral diversity and frequent revisit times make MODIS an invaluable resource for large-scale environmental monitoring and climate research [32].

3.3.3 Data Utilization and Applications

Its data is widely used for various environmental and climate applications. The ability to capture data across multiple spectral bands and at different resolutions allows for comprehensive monitoring and analysis. Key applications of MODIS data include:

- **Land Cover and Land Use:** Supports large-scale land cover mapping and land use classification, which are critical for understanding environmental changes and managing natural resources [22].
- **Vegetation Health:** By monitoring vegetation indices like NDVI (Normalized Difference Vegetation Index), MODIS helps in assessing plant health, drought conditions, and agricultural productivity [32].
- **Climate Studies:** Contributes to climate research by providing data on surface temperatures, cloud properties, and atmospheric composition, essential

for understanding climate dynamics and predicting climate change impacts [22].

- **Disaster Management:** The high temporal resolution of MODIS allows for timely monitoring of natural disasters such as wildfires, floods, and hurricanes, aiding in disaster response and mitigation efforts [32].

3.4 Landsat 8: Technical Details and Trade-offs with Other Missions

3.4.1 Mission Objectives and Launch Information

Landsat 8, the eighth satellite in the Landsat program, was launched on February 11, 2013, from Vandenberg Air Force Base, California. The mission was a collaborative effort between NASA and the United States Geological Survey (USGS) to continue the legacy of the Landsat program, providing critical data for monitoring the Earth's land surface [3].

The primary objectives of Landsat 8 are to capture high-resolution imagery of the Earth's surface, monitor changes over time, and support a wide range of applications, including agriculture, forestry, land use planning, and disaster management. Landsat 8 aims to provide continuous and reliable data to enhance our understanding of natural and human-induced changes on the planet [30].

3.4.2 Instrumentation (OLI and TIRS)

Landsat 8 has two main sensors: the Operational Land Imager (OLI) and the Thermal Infrared Sensor (TIRS).

Operational Land Imager (OLI):

- The sensor captures data in the visible, near-infrared, and shortwave infrared portions of the electromagnetic spectrum.
- It consists of nine spectral bands, including a new deep blue band (coastal/aerosol) and a new cirrus band, which improve the detection of coastal water and high-altitude clouds, respectively.
- The sensor provides higher radiometric resolution, with 12-bit data, enabling the detection of more subtle differences in surface reflectance [3].

Thermal Infrared Sensor (TIRS):

- The sensor captures thermal infrared radiation emitted from the Earth's surface.
- It includes two thermal bands, which are used to measure land surface temperature (LST).
- TIRS was developed to ensure continuity in the thermal data record and to support applications such as monitoring water use and studying heat island effects in urban areas [3].



Figure 3.4: Landsat 8 satellite with OLI and TIRS sensors.

3.4.3 Data Characteristics (Spatial, Spectral, Temporal Resolution)

Landsat 8 provides data with enhanced spatial, spectral, and temporal resolution, making it a valuable tool for detailed Earth observation.

Spatial Resolution:

- The panchromatic band has a spatial resolution of 15 meters.
- The multispectral bands (visible, near-infrared, shortwave infrared) have a spatial resolution of 30 meters.
- The thermal bands have a spatial resolution of 100 meters, resampled to 30 meters [3].

Spectral Resolution:

- Landsat 8 captures data in 11 spectral bands, ranging from the visible to the thermal infrared portions of the spectrum.

- The inclusion of new spectral bands (coastal/aerosol and cirrus) enhances the satellite's ability to monitor specific environmental parameters [3].

Temporal Resolution:

- Landsat 8 has a revisit time of 16 days, ensuring regular and consistent coverage of the Earth's surface.
- This temporal resolution supports the monitoring of dynamic environmental changes and the detection of trends over time [30].

3.4.4 Trade-offs for Land Surface Temperature (LST)

Landsat 8 offers several benefits and trade-offs that make it a powerful tool for measuring Land Surface Temperature (LST). The comparative table 3.1 highlights the trade-offs between Landsat 8, MODIS, and Sentinel-3 for LST measurement.

Attribute	Landsat 8	MODIS (Terra/Aqua)	Sentinel-3
Spatial Resolution	100m (Thermal, resampled to 30m) [3]	1000m [22]	1000m [33]
Temporal Resolution	16 days [30]	1-2 days (global coverage) [22]	1-2 days (global coverage) [33]
Spectral Bands for LST	2 bands (TIRS) [3]	36 bands (includes thermal) [22]	8 bands (includes thermal) [33]
Application Suitability	Urban heat islands, water use, surface temperature monitoring [30]	Large-scale temperature patterns, climate studies, vegetation dynamics [22]	Sea surface temperature, land surface temperature, fire monitoring [33]
Cost and Accessibility	Free, accessible via USGS and Google Earth Engine [3]	Free, accessible via NASA Earthdata [22]	Free, accessible via Copernicus Open Access Hub [33]

Table 3.1: Trade-off between Landsat 8, MODIS, and Sentinel-3 for LST measurement

Landsat 8 is particularly well-suited for detailed LST studies due to its higher

spatial resolution. However, MODIS and Sentinel-3 provide higher temporal resolution, making them ideal for large-scale and frequent temperature monitoring.

3.4.5 Trade-offs for Land Cover Analysis

For land cover analysis, the choice of satellite data depends on the study's specific needs. The comparative table 3.2 highlights the trade-off between Landsat 8, MODIS, and Sentinel-2 for land cover analysis.

Attribute	Landsat 8	MODIS (Terra/Aqua)	Sentinel-2
Spatial Resolution	30m (Multispectral) [3]	250m, 500m, 1000m [22]	10m, 20m, 60m (depending on spectral band) [24]
Temporal Resolution	16 days [30]	1-2 days (global coverage) [22]	5 days [24]
Spectral Bands	11 bands (Visible, Near-Infrared, Shortwave Infrared) [3]	36 bands (Visible, Near-Infrared, Shortwave Infrared, Thermal Infrared) [22]	13 bands (Visible, Near-Infrared, Shortwave Infrared, Red Edge) [24]
Application Suitability	Detailed land cover mapping, agriculture, forestry, urban planning [30]	Large-scale land cover, vegetation dynamics, atmospheric studies [22]	High-resolution land cover mapping, agriculture, forestry, coastal monitoring [24]
Cost and Accessibility	Free, accessible via USGS and Google Earth Engine [3]	Free, accessible via NASA Earthdata [22]	Free, accessible via Copernicus Open Access Hub [24]

Table 3.2: Trade-off between Landsat 8, MODIS, and Sentinel-2 for land cover analysis

MODIS is particularly well-suited for land cover analysis in Google Earth Engine (GEE) due to its high temporal resolution, large coverage area, and spectral diversity. Landsat 8 and Sentinel-2, with their higher spatial resolution, are better suited for detailed local studies.

3.5 Summary of Landsat and MODIS Data Utilization

The Landsat and MODIS programs, with their respective histories and continuous improvements, have established themselves as cornerstones in the field of Earth observation. The comprehensive data these satellite programs provide is crucial in environmental monitoring, resource management, and scientific research.

The detailed imagery and thermal data captured by Landsat 8's OLI and TIRS sensors enable the detection and analysis of various environmental phenomena, including urban heat islands, deforestation, and agricultural practices. MODIS's high temporal resolution and extensive spectral coverage support large-scale environmental monitoring and climate research, providing valuable insights into land cover dynamics and atmospheric conditions.

Despite the trade-offs in spatial and temporal resolutions, the benefits of Landsat 8 and MODIS, such as improved radiometric resolution, high temporal frequency, and data accessibility, make them invaluable resources for a wide range of users. The data's accessibility through platforms like Google Earth Engine further enhances its utility, enabling efficient data processing and analysis for various applications.

Overall, the Landsat and MODIS programs continue to provide essential data that supports critical environmental monitoring efforts, contributing to our understanding and management of the Earth's resources.

Chapter 4

Technical Approach of Heat Island Detection and Mitigation with Land Cover Insights

Traditional methods for measuring Heat Islands involve using physical weather stations or multiple thermometers placed across different locations. While these methods are accurate, they can be resource-intensive and offer limited spatial coverage. The emergence of remote sensing technology has made it possible to utilize satellite data to measure surface temperatures across large regions, offering a detailed perspective on the Heat Island effect. [3, 15].

This chapter explores a generic algorithm that utilizes satellite data for Heat Island detection. It describes the methods for processing this data, defining the area of interest, and developing an application to detect Heat Islands, including their hot and cold spots. This approach leverages the capabilities of Google Earth Engine (GEE) and Landsat 8 satellite data to provide a scalable and efficient solution for urban climate analysis.

4.1 Land Surface Temperature Data

Land Surface Temperature (LST) data is crucial for analyzing Heat Islands. This application uses data from the Landsat 8 satellite to provide surface temperature measurements. Landsat 8 offers high-resolution thermal data, which is essential for detecting temperature variations across urban and rural areas. The process involves filtering the data by location, date range, and cloud cover to ensure high-quality temperature readings. Satellite data allows for consistent and repeatable measurements, making it ideal for long-term climate studies [10, 16].

Google Earth Engine (GEE) is a cloud-based platform that facilitates the analysis

and visualization of geospatial data. It hosts petabytes of satellite imagery and other geospatial datasets, allowing users to perform complex analyses without needing local storage or significant computational power [3, 15]. The use of GEE enables efficient processing and analysis of large datasets, such as those from Landsat 8.

First, the data is filtered to exclude clouds using a cloud masking function. Cloud cover can significantly affect the accuracy of temperature measurements. Therefore, the cloud masking function 4.1 is applied to remove pixels contaminated by clouds.

$$\text{mask} = \text{qa.bitwiseAnd}(1 << 3).\text{Or}(\text{qa.bitwiseAnd}(1 << 4)) \tag{4.1}$$

This function works by selecting the 'QA_PIXEL' band, which contains quality assessment information. It uses bitwise operations to identify cloud and shadow pixels and then masks them out, ensuring that only clear-sky pixels are used in subsequent analysis. The QA_PIXEL band is part of the Landsat 8 data product and provides a reliable way to filter out cloud-covered areas, which is crucial for obtaining accurate surface temperature readings [3].

Next, scale factors are applied to convert the thermal band data from the Landsat 8 satellite into surface temperature values in Celsius. Landsat 8's thermal infrared sensor measures radiance, which needs to be converted to temperature as shown in the 4.2 equation.

$$T_{\text{Celsius}} = (L \times 0.00341802) + 149.0 - 273.15 \tag{4.2}$$

where L is the radiance value from the thermal band.

This step is essential for ensuring that the temperature data is in a usable format for further analysis and comparison [5].

After processing, the 4.3 equation shows that the data is averaged over the specified time period to produce a mean surface temperature image for the area of interest.

$$T_{\text{mean}} = \frac{1}{n} \sum_{i=1}^{n} T_i \tag{4.3}$$

This step is crucial as it reduces the impact of any anomalous temperature readings and provides a stable and reliable measure of surface temperature for the analysis. Averaging the data over a specified period helps mitigate the effects of short-term temperature fluctuations and provides a clearer picture of the overall temperature trends [16].

Figure 4.1 illustrates the distribution of surface temperatures across the study area, highlighting the variation in land surface temperature data obtained from Landsat 8.

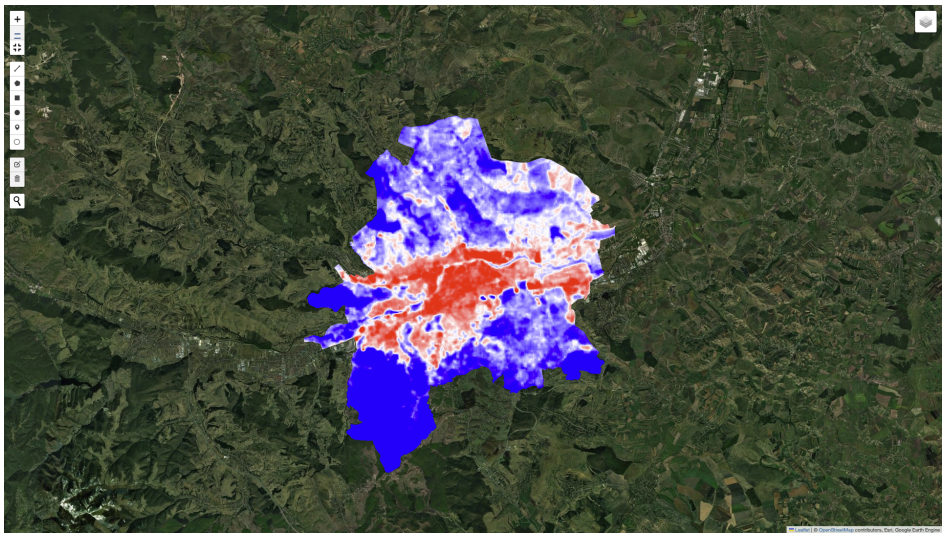


Figure 4.1: Surface Temperature Map

Using Google Earth Engine for this process offers several advantages:

- **Scalability**: GEE can handle large datasets and complex computations, making it suitable for analyzing extensive geographical areas over long time periods.
- **Access to Historical Data**: GEE provides access to historical Landsat data, allowing for analyzing temporal changes in surface temperature.
- **Integration with Other Datasets**: GEE can integrate various geospatial datasets, facilitating comprehensive environmental and climate studies.

The use of LST data from Landsat 8, combined with the processing power and data accessibility of Google Earth Engine, forms a robust foundation for detecting and analyzing Heat Islands.

4.2 Heat Island Analysis

Detecting Heat Islands involves identifying areas where the surface temperature significantly exceeds the mean temperature of the region. This is achieved by calculating the mean and standard deviation of the temperatures within the Area of Interest (AOI). These statistical measures set dynamic thresholds to identify and classify different temperature zones, including heat islands, hot spots, and cold spots. This method provides a robust way to detect significant temperature anomalies within urban environments [17, 4].

4.2.1 Calculating Mean and Standard Deviation

The first step in the analysis is to compute the mean and standard deviation of the surface temperature within the AOI. These statistics provide a baseline for defining what constitutes a heat island or other temperature anomalies:

$$\mu = \frac{1}{N} \sum_{i=1}^{N} x_i \tag{4.4}$$

$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^{N} (x_i - \mu)^2} \tag{4.5}$$

where μ is the mean temperature, σ is the standard deviation, x_i are the temperature values, and N is the number of pixels in the AOI.

4.2.2 Setting Dynamic Thresholds

Based on the calculated mean and standard deviation, dynamic thresholds are set to classify different temperature zones:

$$T_{\text{heat island}} = \mu + \sigma \tag{4.6}$$

$$T_{\text{hot spot}} = \mu + 2\sigma \tag{4.7}$$

$$T_{\text{cold spot}} = \mu - \sigma \tag{4.8}$$

These thresholds define the boundaries for heat islands and hot and cold spots.

4.2.3 Detecting Heat Islands

To identify and unify heat islands, morphological operations are applied. The core idea is to create a binary mask where pixels exceeding the heat island threshold are set to 1. This mask is then processed using the `focal_max` function, which expands the identified areas by considering neighboring pixels. The `focal_max` function applies a circular kernel over the binary mask. The radius of this kernel and the number of iterations define how much the identified areas are expanded.

$$\text{connected} = \text{mask.focal_max}(\text{kernel} = \text{circle}(\text{radius} = 7), \text{iterations} = 8) \tag{4.9}$$

In this equation, `connected` represents unified heat island areas. The `mask` is generated by thresholding the temperature image, where pixels with temperatures

greater than the heat island threshold are set to 1. The `focal_max` function expands these areas, effectively merging smaller clusters into larger contiguous regions.

The function 4.1 encapsulates this process. The `mask` is created by applying a threshold to the surface temperature image. Pixels with temperatures above the heat island threshold are set to 1, while others are set to 0. This binary mask is then expanded using the `focal_max` function, which considers neighboring pixels within a circular radius of 7 pixels over 8 iterations. This process ensures that smaller heat islands merge into larger, more significant heat island regions.

```
def detect_heat_island_contours(image, threshold_temp):
    mask = image.gt(threshold_temp).selfMask()
    connected = mask.focal_max(kernel=ee.Kernel.circle(radius
       =7), iterations=8)
    vectors = connected.reduceToVectors(
        geometryType='polygon',
        reducer=ee.Reducer.countEvery(),
        scale=30,
        maxPixels=1e8,
        geometry=aoi
    )
    return vectors
```

Listing 4.1: `detect_heat_island_contours` function

Finally, the `reduceToVectors` function is used to convert these connected regions into vector polygons. These vector polygons are essential for further analysis and visualization, as they provide a clear and interpretable representation of the heat islands within the area of interest (AOI).

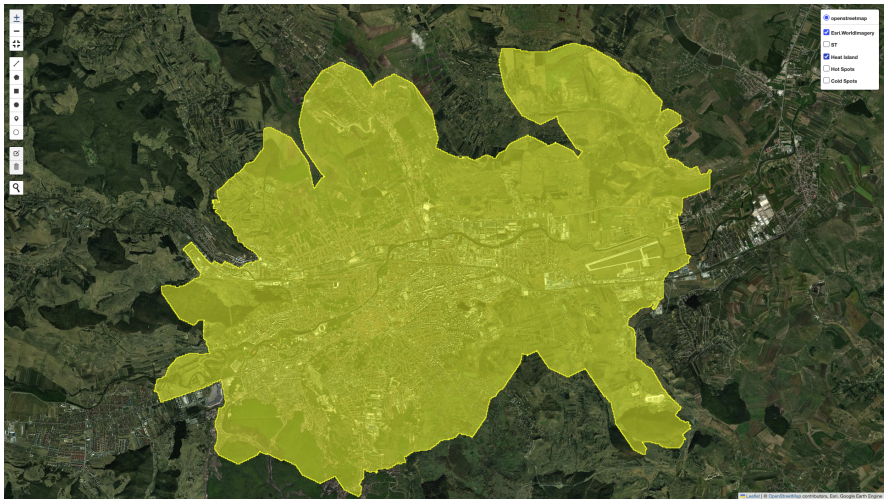


Figure 4.2: Heat Island Map

The detected heat islands are shown in Figure 4.2, clearly indicating areas with significantly higher temperatures than the surrounding regions.

4.2.4 Detecting Hot Spots and Cold Spots

Hot spots and cold spots are detected using a similar approach but with different parameters to accurately identify these regions. Hot spots are areas where the temperature exceeds the mean by at least two standard deviations, indicating extreme heat. A smaller kernel radius and fewer iterations are used to identify hot spots to ensure that only the most intense areas are highlighted.

$$\text{connected_hot_spot} = \text{mask.focal_max}(\text{kernel} = \text{circle}(\text{radius} = 1), \text{iterations} = 1) \tag{4.10}$$

The 4.2 function operates similarly to the heat island detection function but with adjusted parameters. The radius of the kernel is set to 1 pixel, and the number of iterations is reduced to 1. This ensures that only the most intense hot spots are identified without unnecessarily merging them into larger areas. The binary mask is created by thresholding the temperature image where pixels with temperatures above the hot spot threshold are set to 1. The `focal_max` function then expands these areas slightly to include neighboring pixels, highlighting only the most intense hot spots.

```
def detect_hot_spot_contours(image, threshold_temp):
    mask = image.gt(threshold_temp).selfMask()
    connected = mask.focal_max(kernel=ee.Kernel.circle(radius
       =1), iterations=1)
    vectors = connected.reduceToVectors(
        geometryType='polygon',
        reducer=ee.Reducer.countEvery(),
        scale=30,
        maxPixels=1e8,
        geometry=aoi
    )
    return vectors
```

Listing 4.2: `detect_hot_spot_contours` function

Within the heat islands, specific hot spots are identified as shown in Figure 4.3. These are areas with the highest temperature concentrations.

Cold spots, conversely, are areas where the temperature is significantly lower

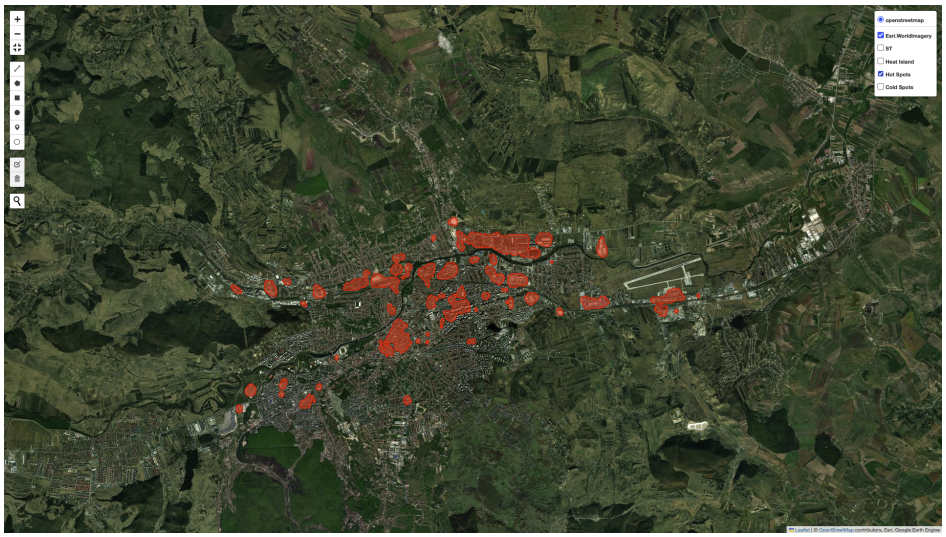


Figure 4.3: Hot Spot Map

than the mean by at least one standard deviation. The `focal_min` function is used here to contract the areas, effectively merging smaller cold spots into larger regions.

$$\text{connected_cold_spot} = \text{mask.focal_min}(\text{kernel} = \text{circle}(\text{radius} = 2), \text{iterations} = 3) \tag{4.11}$$

The 4.3 function identifies these areas using a radius of 2 pixels and 3 iterations. The binary mask is created by thresholding the temperature image where pixels with temperatures below the cold spot threshold are set to 1. The `focal_min` function contracts these areas by considering neighboring pixels within a circular radius of 2 pixels over 3 iterations, ensuring that even the smallest cold spots are included in the analysis.

```
def detect_cold_spot_contours(image, threshold_temp):
    mask = image.lt(threshold_temp).selfMask()
    connected = mask.focal_min(kernel=ee.Kernel.circle(radius
       =2), iterations=3)
    vectors = connected.reduceToVectors(
        geometryType='polygon',
        reducer=ee.Reducer.countEvery(),
        scale=30,
        maxPixels=1e8,
        geometry=aoi
    )
    return vectors
```

Listing 4.3: `detect_cold_spot_contours` function

Conversely, Figure 4.4 presents the cold spots, areas with lower temperatures that act as cooling zones within the urban landscape.

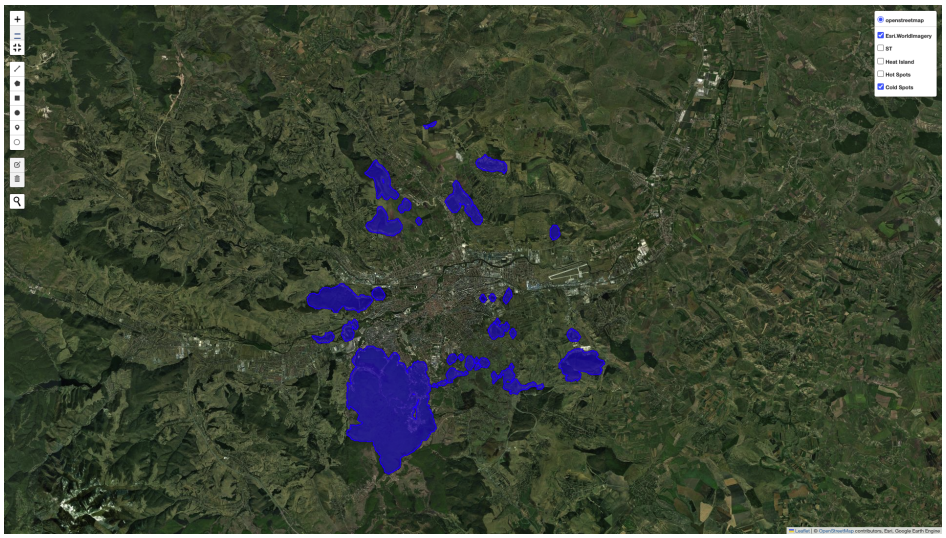


Figure 4.4: Cold Spot Map

The use of dynamic thresholds and morphological operations in Google Earth Engine enables the efficient detection and classification of temperature anomalies such as heat islands, hot spots, and cold spots. By adjusting the kernel radius and iteration parameters, the algorithm can accurately identify and classify temperature extremes within urban environments, providing a robust framework for urban climate analysis. This method allows for precisely delineating areas with significant temperature anomalies, facilitating targeted interventions and mitigation strategies.

4.3 Land Cover Analysis

Land cover analysis provides insights into the composition of urban and rural areas, which is crucial for understanding the factors contributing to Heat Islands. The analysis uses MODIS Land Cover Type data to classify different land cover types. Understanding the distribution of various land cover types helps assess how different surfaces contribute to the Urban Heat Island (UHI) effect [9, 6].

4.3.1 Data Processing

The first step in the land cover analysis involves defining the Area of Interest (AOI) based on the coordinates retrieved from OpenStreetMap. This AOI is then used to clip the land cover data from the MODIS Land Cover Type dataset. The MODIS dataset provides annual land cover data at a 500m resolution, which is sufficient for broad-scale urban analysis.

Once the AOI is defined, the land cover data is clipped to this region using Google Earth Engine (GEE). The `clip` function in GEE ensures that only the pixels within the AOI are considered for further analysis. This is crucial for analyzing the specific urban or peri-urban area of interest.

```
landcover = ee.Image('MODIS/006/MCD12Q1/2019_01_01').select('
    LC_Type1')
landcover_aoi = landcover.clip(aoi)
```

Listing 4.4: GEE MODIS data retreival

After clipping the data, the next step is to calculate the area occupied by each land cover type within the AOI. This involves multiplying each pixel by its area (500m x 500m) and summing these values for each land cover type. The total area for each type is then used to calculate the percentage of the AOI covered by each land cover type.

$$\text{area} = \sum \text{pixel_area} \tag{4.12}$$

The formula above represents the total area covered by each land cover type, where `pixel_area` is the area of a single pixel (250,000 m^2 for a 500m MODIS pixel). The `reduceRegion` function in GEE, combined with a sum reducer, facilitates this calculation.

```
areaImage = ee.Image.pixelArea().addBands(landcover_aoij)
areaReducer = ee.Reducer.sum().group(1, 'landcover_type')

landcoverArea = areaImage.reduceRegion(
    reducer=areaReducer,
    geometry=aoi,
    scale=500,
    maxPixels=1e8
)
```

Listing 4.5: `landcoverArea` function

This code snippet calculates the total area for each land cover type. The `pixelArea` function creates an image where each pixel value represents its area. This is then combined with the clipped land cover data to create a multi-band image. The `reduceRegion` function with a group reducer is used to sum the areas of pixels for each land cover type.

The calculated areas are then converted into percentages by dividing with the total AOI area. This step normalizes the data, making comparing the relative proportions of different land cover types easier.

4.3.2 Land Cover Categories

The land cover types provided by MODIS are detailed and specific, but these types are often grouped into broader categories for practical analysis. This helps in understanding the overall composition of the landscape and its impact on urban climate.

The land cover types are mapped to broader categories such as Dense Green Area, Water Area, Urban Built-up Area, Agriculture Area, and Mixed Green, Water, and Built-up Area. This categorization simplifies the analysis and helps identify the dominant land cover types contributing to the UHI effect.

```
landcover_names = {
    1: 'Evergreen Needleleaf forest',
    2: 'Evergreen Broadleaf forest',
    3: 'Deciduous Needleleaf forest',
    4: 'Deciduous Broadleaf forest',
    5: 'Mixed forest',
    6: 'Closed shrublands',
    7: 'Open shrublands',
    8: 'Woody savannas',
    9: 'Savannas',
    10: 'Grasslands',
    11: 'Permanent wetlands',
    12: 'Croplands',
    13: 'Urban and built-up',
    14: 'Cropland/Natural vegetation mosaic',
    15: 'Snow and ice',
    16: 'Barren or sparsely vegetated',
    17: 'Water Bodies'
}

category_mapping = {
    'Dense green area': [1, 2, 3, 4, 5, 6, 7, 8, 9],
    'Water Area': [11, 17, 15],
    'Urban built-up area': [13],
    'Agriculture area': [12, 14],
    'Mixed green, water and built-up area': [10, 16]
}
```

Listing 4.6: Land Cover Category Classification

The `category_mapping` dictionary groups the specific MODIS land cover types into broader categories. This grouping is essential for understanding how different

land cover types contribute to the UHI effect. For instance, urban built-up areas are expected to significantly impact increasing local temperatures due to their high heat absorption and low albedo.

The percentage of each category is calculated by summing the areas of the land cover types within each category and then normalizing these values by the total AOI area. This provides a clear picture of the composition of the landscape and highlights the dominant factors contributing to the UHI effect.

By analyzing the distribution of these categories within the AOI, we can identify potential areas for urban planning interventions. For instance, increasing the proportion of green areas or water bodies can help mitigate the UHI effect by enhancing evapotranspiration and providing cooling effects through shading [2, 7].

Overall, land cover analysis using MODIS data and GEE provides a robust framework for understanding the landscape composition and its impact on urban climate. This analysis is crucial for developing effective strategies to mitigate the UHI effect and improve urban livability.

4.4 OpenStreetMap Boundary Retrieval

Defining the Area of Interest (AOI) is crucial for focused analysis. Using the OpenStreetMap API, the AOI is determined based on the city name and address type. This approach leverages the detailed geographic data provided by OpenStreetMap to accurately define the boundaries of the area being analyzed [34].

The first step is to query the OpenStreetMap API to retrieve the bounding box coordinates for the specified city and address type. This is done using the `requests` library in Python, which allows for sending HTTP requests and handling responses. The constructed URL includes parameters such as the city name and request format to ensure accurate data retrieval 4.7.

```
1 url = f'https://nominatim.openstreetmap.org/search.php?q={
    city_name}&polygon_geojson=1&format=json'
2 response = requests.get(url, headers={'User-Agent': 'Mozilla
    /5.0'})
3 data = response.json()
```

Listing 4.7: OpenStreetMap URL Querry

The response from the OpenStreetMap API contains geographic data in JSON format. The data is processed to extract the bounding box coordinates. The JSON response is parsed to find the appropriate bounding box based on the specified address type 4.8.

```
for city_data in data:
    if city_data.get('addresstype') == type:
        bounding_box = city_data.get('geojson').get('
           coordinates')
```

Listing 4.8: OpenStreetMap GeoJSON response

Depending on the geometry type, which can be either `Polygon` or `MultiPolygon`, the function will return the appropriate coordinates from the bounding box structure. This distinction is important as it ensures the correct interpretation and handling of the geographic data provided.

```
if city_data.get('geojson').get('type') == 'MultiPolygon':
    return bounding_box[0][0]
elif city_data.get('geojson').get('type') == 'Polygon':
    return bounding_box[0]
```

Listing 4.9: OpenStreetMap GeoJSON Processing

The code 4.9 first checks the type of geometry in the `geojson` field of the `city_data` dictionary. If the type is `MultiPolygon`, it returns the first coordinate of the first polygon in the bounding box. If the type is `Polygon`, it returns the first coordinate of the bounding box.

Once the bounding box is retrieved, it is converted into an Earth Engine geometry object. This is done using the function 4.10, which calls `get_bounding_boxes` and converts the bounding box coordinates into an Earth Engine `Polygon` geometry object if the bounding box is not `None`:

```
def get_aoi(city_name, address_type):
    bounding_box = get_bounding_boxes(city_name, address_type
       )
    if bounding_box:
        return ee.Geometry.Polygon([bounding_box])
    else:
        return None
```

Listing 4.10: `get_aoi` function

The AOI defined by the bounding box is crucial for focusing the analysis on the specific urban area of interest. This ensures that the data processing and analysis steps that follow are accurately targeted to the relevant geographical region, providing precise and relevant results for heat island detection and other analyses.

Using the bounding box coordinates from OpenStreetMap allows for flexibility and precision in defining the AOI, making the methodology adaptable to different cities and administrative divisions. This approach is essential for urban climate studies, where the accuracy and relevance of spatial data are paramount [34].

4.5 Parallelization and Efficient Processing

To improve efficiency, tasks such as contour detection and CSV writing are executed in parallel using Python's `concurrent.futures` module. This approach leverages multi-threading to perform simultaneous computations, significantly reducing the overall processing time and enhancing the performance of the analysis pipeline.

Parallelization is crucial when dealing with large datasets and computationally intensive tasks, such as heat island contour detection and data export. Distributing these tasks across multiple threads allows the system to utilize computational resources more effectively, leading to faster execution and improved response.

The parallel execution is managed using a thread pool executor, which allows for the concurrent execution of functions. The `Executor.submit(task)` method is used to submit tasks for parallel execution. This is illustrated in the following formula, where multiple tasks are submitted to the executor:

$$\text{Executor.submit(task)} \tag{4.13}$$

For example, in the heat island detection script, parallelization is applied to the contour detection functions for heat islands, hot spots, and cold spots in code sequence 4.11.

```
1 with concurrent.futures.ThreadPoolExecutor() as executor:
2     heat_island_future = executor.submit(
        detect_heat_island_contours, clip_mean_ST.select('
        ST_B10'), heat_island_threshold_temp)
3     hot_spot_future = executor.submit(
        detect_hot_spot_contours, clip_mean_ST.select('ST_B10'
        ), hot_spot_threshold_temp)
4     cold_spot_future = executor.submit(
        detect_cold_spot_contours, clip_mean_ST.select('ST_B10
        '), cold_spot_threshold_temp)
5
6     heat_island_contours = heat_island_future.result()
7     hot_spot_contours = hot_spot_future.result()
8     cold_spot_contours = cold_spot_future.result()
```

Listing 4.11: Paralel execution of contours detection

- A thread pool executor is created using `concurrent.futures.ThreadPoolExecutor()`.
- The `submit` method is used to schedule the 4.1, 4.2 and 4.3 functions for parallel execution.
- The `result` method is called on each future object to retrieve the results once the tasks are completed.

Similarly, parallelization is employed when writing the output data to CSV files. This ensures that the data export operations do not bottleneck the overall process in the code snippet 4.12.

```
with concurrent.futures.ThreadPoolExecutor() as executor:
    html_future = executor.submit(write_html)
    boundary_csv_future = executor.submit(write_boundary_csv)
    stats_csv_future = executor.submit(write_stats_csv)

    html_file_path = html_future.result()
    boundary_csv_path = boundary_csv_future.result()
    stats_csv_path = stats_csv_future.result()
```

Listing 4.12: Paralel execution of CSV files writing

The parallelization strategy in this project follows best practices for efficient processing in data-intensive applications. By leveraging multi-threading, the system achieves significant performance gains, enabling faster and more scalable analysis of urban heat islands.

This approach is essential for handling large-scale geospatial data and computationally demanding tasks, ensuring timely and accurate results. Parallelization improves processing speed and enhances the user experience by reducing the time required to generate and display results [35].

4.6 PDF Generation and GPT-based Recommendations

The analysis results are compiled into a PDF report, which includes land cover percentages, heat island statistics, and recommendations generated by GPT. The PDF report is generated using the `fpdf` library, a Python library for creating PDF documents. Including GPT-3 generated insights adds a layer of expert recommendations to the data-driven analysis.

4.6.1 PDF Generation

The `fpdf` library is utilized to structure the PDF report. This involves creating a header, adding sections for analysis results, and formatting the content for readability. The process begins with setting up the document layout and styles in class 4.13.

```
from fpdf import FPDF

class PDF(FPDF):
    def header(self):
        self.set_font("Arial", 'B', 10)
        self.cell(0, 10, f"{city_name} Analysis for Land
            Cover and Heat Islands", 0, 1, 'C')
        self.image(logo_path, x=self.w - 30, y=8, w=20)
        self.ln(10)
```

Listing 4.13: PDF class definition

In the PDF generation process, sections are added for different parts of the analysis, including heat island statistics and land cover percentages. This structured approach ensures that the PDF report is comprehensive and professional, providing a detailed overview of the analysis results.

4.6.2 GPT-based Recommendations

Recommendations for mitigating urban heat effects are generated using OpenAI's GPT-3. The GPT-3 model processes the input prompt, which includes land cover and heat island statistics, to offer actionable insights and suggestions for urban heat mitigation. This is particularly valuable for urban planners and policymakers seeking expert advice based on the analysis.

The prompt is crafted to include relevant data and request specific types of recommendations:

"[City Name] has the following land cover percentage: [Land Cover Percentages] and the following heat island statistics: [Heat Island Statistics]. Create a list of possible solutions for reducing the temperature in [City Name]. Give some advice to the citizens and also some ideas for the administrative council to help reduce and prevent the urban heat island."

The GPT-3 model then generates a detailed response based on the provided data. This response includes practical suggestions for both citizens and the administrative council to mitigate the effects of urban heat islands. The generated recommendations are included in the PDF report, providing a blend of data-driven analysis and expert suggestions.

The combination of data analysis, PDF reporting, and AI-generated recommendations represents a robust approach to addressing urban heat islands. This methodology ensures that the results are not only accurate and detailed but also actionable and informative, aiding in the development of effective urban heat mitigation strategies [36].

4.7 Conclusion

Integrating advanced geospatial analysis, remote sensing data, and AI-driven insights provides a comprehensive framework for urban heat island detection and analysis. Utilizing Google Earth Engine (GEE) for extensive geospatial dataset processing, we achieve accurate identification and classification of temperature anomalies through dynamic thresholding and morphological operations.

The foundation of this analysis is high-quality Landsat 8 data, processed to derive Land Surface Temperature (LST). Cloud masking and scale factor adjustments ensure accurate temperature readings. Statistical approaches using mean and standard deviation facilitate adaptable dynamic thresholds for various urban environments. MODIS Land Cover Type data adds depth to the analysis, identifying land cover patterns affecting heat islands. Classifying land cover types and calculating their areas and percentages within the Area of Interest (AOI) reveal contributions to heat retention. AI-driven recommendations by OpenAI's GPT-3 enhance practical utility, offering actionable insights for urban heat mitigation tailored for citizens and planners. These insights, integrated into PDF reports, ensure informative and actionable findings. Efficiency is improved with parallel processing using Python's `concurrent.futures` module, reducing computation time. The web application, built with React and Spring Boot, ensures robust performance and scalability and is deployed on AWS Cloud Infrastructure for reliability and security. Access to detailed land cover analysis and PDF generation is restricted to logged-in users, providing personalized and secure insights for urban planners and researchers, and maintaining data security and privacy.

In summary, this methodology offers a powerful tool for understanding and addressing urban heat islands. Combining geospatial analysis, remote sensing data, and AI-driven insights supports effective strategies for mitigating heat island effects, contributing to sustainable urban environments. Future work will enhance the system's capabilities by integrating real-time data streams and more sophisticated AI models. These improvements will refine heat island detection and mitigation strategies, making urban climate analysis more dynamic and responsive to changing conditions. By continuously evolving the framework, we aim to provide cities with tools to adapt to and combat challenges posed by urban heat islands [3, 21, 36].

Chapter 5

Application Development

5.1 Architecture

5.1.1 Overview

The architecture of the Heat Island Analysis Tool is designed to efficiently process, analyze, and visualize urban heat island data. The system integrates various components, including data sources, preprocessing scripts, backend services, interfaces, and monitoring tools. The overall architecture is depicted in Figure 5.1.

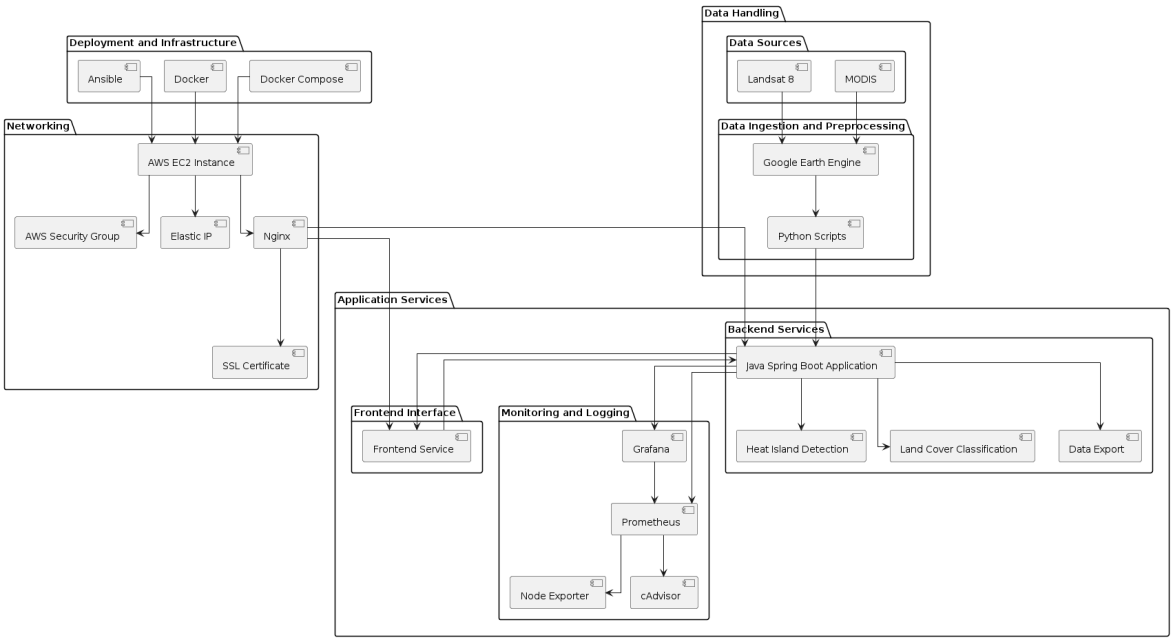


Figure 5.1: High-Level Architecture of the Heat Island Analysis Tool

The architecture consists of the following main packages:

Deployment and Infrastructure This package includes the tools and configurations used for deploying the application:

- **Ansible**: Used for automating the application deployment.
- **Docker**: Containerizes the application for consistent runtime environments.
- **Docker Compose**: Orchestrates multi-container Docker applications.

Networking This package handles the network configuration and security:

- **AWS EC2 Instance**: Hosts the application.
- **AWS Security Group**: Defines the security rules for the instance.
- **Elastic IP**: Provides a static IP address for the instance.
- **Nginx**: Acts as a reverse proxy and handles SSL termination.
- **SSL Certificate**: Ensures secure communication over HTTPS.

Application Services
This package contains the core services of the application:

- **Backend Services**:
  - **Java Spring Boot Application**: The main backend framework.
  - **Heat Island Detection**: Processes satellite data to detect heat islands.
  - **Land Cover Classification**: Analyzes land cover data.
  - **Data Export**: Handles exporting of processed data.
- **Frontend Interface**:
  - **Frontend Service**: Provides the user interface for interacting with the tool.
- **Monitoring and Logging**:
  - **Grafana**: Visualizes monitoring data.
  - **Prometheus**: Collects and stores metrics.
  - **Node Exporter**: Collects hardware and OS metrics from the machine running the containers.
  - **cAdvisor**: Collects container-level metrics and sends them to Prometheus for storage and analysis.

Data Handling
This package is responsible for data acquisition and preprocessing:

- **Data Sources**:
  - **Landsat 8**: Provides high-resolution satellite imagery.
  - **MODIS**: Offers moderate-resolution imaging for large-scale phenomena.
- **Data Ingestion and Preprocessing**:
  - **Google Earth Engine**: Facilitates planetary-scale geospatial analysis [21].
  - **Python Scripts**: Preprocesses and prepares data for analysis.

This modular architecture ensures that each component can be developed, deployed, and scaled independently, facilitating robust and efficient analysis of urban heat island phenomena.

5.1.2 Backend Services

The backend services of the Heat Island Analysis Tool are developed using Java Spring Boot, which provides a solid and scalable framework for managing the application's core functionalities. These services include various controllers and services essential for handling user interactions, data processing, and authentication. The architecture of the backend services is depicted in Figure 5.2.

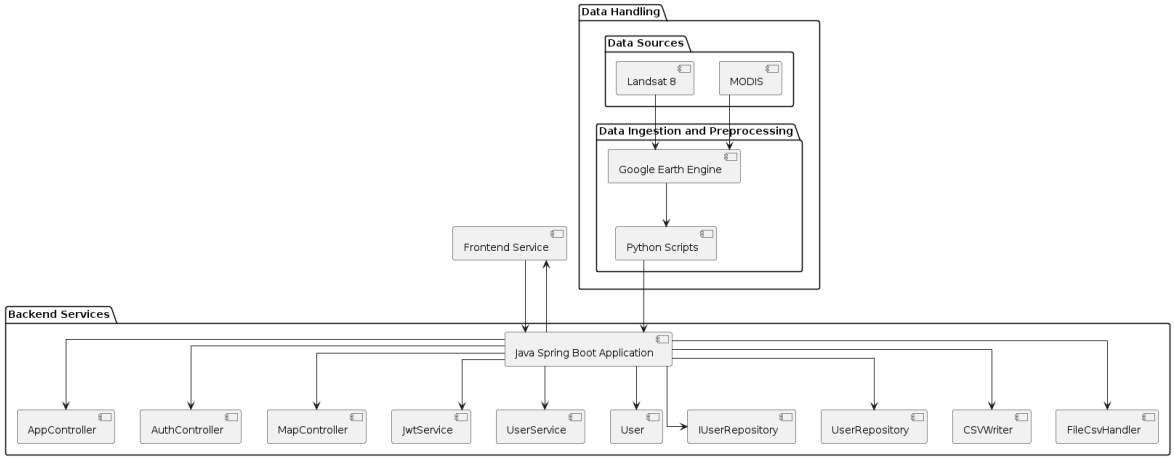


Figure 5.2: Architecture of the Backend Services

The backend services consist of the following key components:

- **Java Spring Boot Application**: Serves as the main framework for the backend, orchestrating various services and managing the overall workflow.
- **AppController**: Manages general application-level requests and interactions.

- **AuthController**: Handles user authentication, including login and token generation.
- **MapController**: Manages requests related to heat island and map generation, including data processing triggers and file handling.
- **JwtService**: Provides functionalities for generating and validating JWT tokens, ensuring secure user sessions.
- **UserService**: Manages user-related operations, such as user creation, retrieval, and authentication.
- **User**: Represents the user entity in the application.
- **IUserRepository and UserRepository**: Interfaces and implementations for accessing user data from the database.
- **CSVWriter and FileCsvHandler**: Utilities for handling CSV file operations, particularly for data export.
- **Frontend Service**: Handles the interaction with the user interface, managing requests from the frontend container.

The backend services rely on external data sources and preprocessing scripts to acquire and prepare data for analysis:

- **Data Sources**:
  - **Landsat 8**: Provides high-resolution satellite imagery, crucial for detailed analysis of urban heat islands.
  - **MODIS**: Offers moderate-resolution imaging, suitable for large-scale environmental monitoring and analysis.
- **Data Ingestion and Preprocessing**:
  - **Google Earth Engine**: Facilitates planetary-scale geospatial analysis, enabling the processing of vast amounts of satellite data [21].
  - **Python Scripts**: Preprocess and prepare data for analysis, ensuring the data is in the required format for further processing.

The modular design of the backend services ensures each component can be developed, deployed, and scaled independently, providing a flexible and efficient architecture for urban heat island analysis.

By leveraging Java Spring Boot's robust ecosystem, the backend services can handle complex data processing workflows, secure user authentication, and efficient data management, ensuring a reliable and scalable solution for urban climate analysis.

5.1.3 Frontend Interface

The frontend interface of the Heat Island Analysis Tool is designed to provide a seamless user experience for interacting with the tool. The frontend is built using React and is containerized for consistent deployment across different environments. The architecture of the frontend interface is depicted in Figure 5.3.

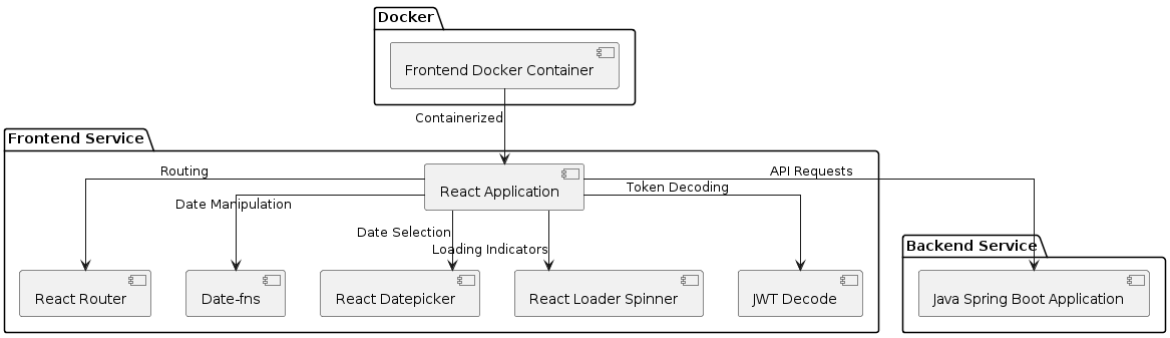


Figure 5.3: Architecture of the Frontend Interface

The frontend interface consists of the following key components:

- **Login Form**: Facilitates user authentication by interacting with the backend's authentication service. It uses React components, React Router, and JWT decoding to manage user sessions.
- **Main Form**: Allows users to input data for heat island analysis, including location, date range, and type of area. Utilizes React components such as DatePicker and form validation with Date-Fns.
- **Data Display**: Shows the heat map and statistical data retrieved from the backend after form submission. Includes loaders from React Loader Spinner to indicate data fetching status.
- **Protected Routes**: Ensures that only authenticated users can access certain parts of the application, using React Router and context for managing the authentication state.

The integration of these components ensures that the frontend interface is responsive, user-friendly, and capable of efficiently interacting with the backend services to provide real-time data and analysis to users.

5.1.4 Data Handling

The data handling component of the Heat Island Analysis Tool acquires, preprocesses and manages data for heat island detection and land cover analysis. It integrates various data sources, including satellite imagery and geospatial datasets, with preprocessing scripts to ensure data readiness. Figure 5.4 shows the architecture of this component.

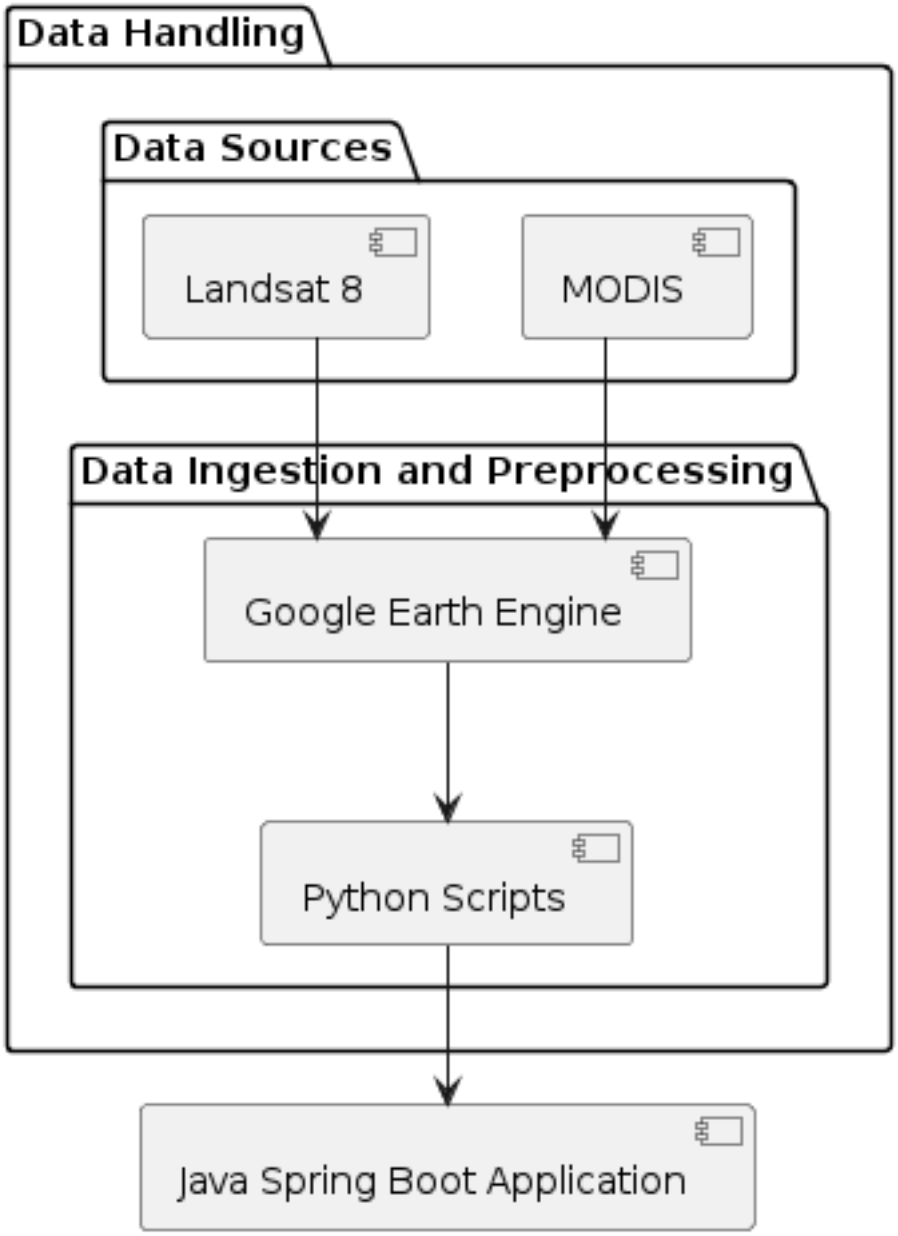


Figure 5.4: Architecture of the Data Handling Component

The data handling package includes:

- **Data Sources**:
  - **Landsat 8**: Provides high-resolution satellite imagery for detailed urban heat island analysis [31].
  - **MODIS**: Offers moderate-resolution imaging for large-scale environmental monitoring [22].
- **Data Ingestion and Preprocessing**:
  - **Google Earth Engine**: Enables planetary-scale geospatial analysis, processing vast amounts of satellite data [21].
  - **Python Scripts**: Preprocess and prepare data, ensuring it is in the required format for analysis [5].

These integrated data sources and preprocessing tools ensure the data is accurate, high-quality, and ready for analysis and visualization.

5.1.5 Monitoring and Logging

The monitoring and logging component ensures the application runs smoothly by tracking performance and logging events. It uses Prometheus for metrics collection and Grafana for visualization. Figure 5.5 shows its architecture.

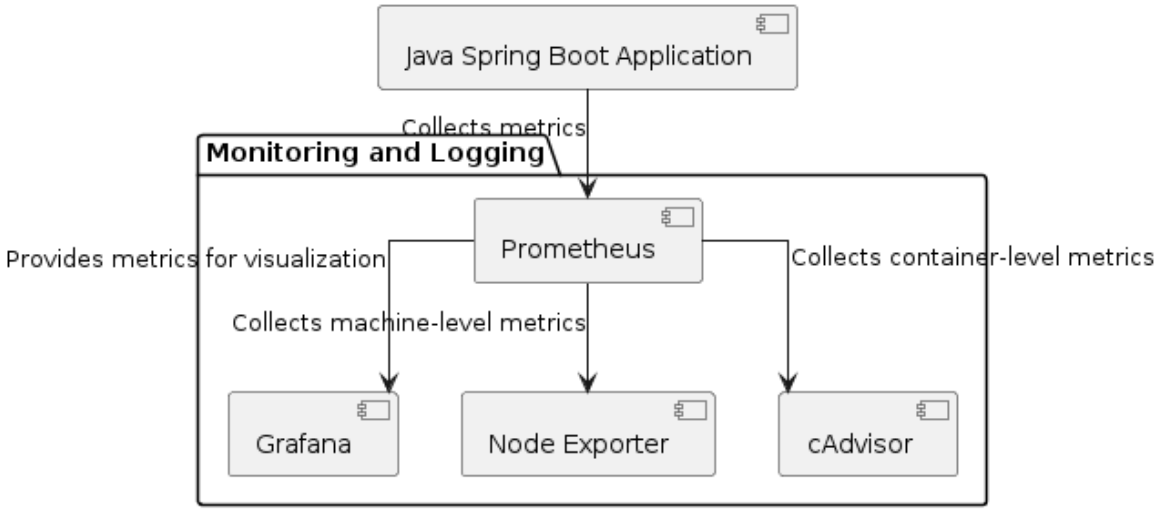


Figure 5.5: Architecture of the Monitoring and Logging Component

The monitoring and logging package includes:

- **Prometheus**: Collects and stores metrics from the application, infrastructure, and Node Exporter for machine-level metrics [37].
- **Grafana**: Visualizes collected metrics, providing real-time insights into performance and health [37].
- **cAdvisor**: Collects container-level metrics for Prometheus.
- **Java Spring Boot Application**: Backend services expose metrics for Prometheus.
- **Node Exporter**: Collects hardware and OS metrics from the machine running the containers.

Grafana dashboards include:

- **Container Logs**: Displays container logs to monitor activities and troubleshoot issues.
- **Container Stats**: Shows key metrics like CPU usage, memory usage, network I/O, and uptime for each container.
- **Machine Stats**: Displays the hosting machine's overall CPU and memory usage.

By leveraging Prometheus, cAdvisor, and Grafana, this component provides comprehensive performance tracking and quick issue identification, maintaining the Heat Island Analysis Tool's reliability and efficiency.

Integrating these monitoring tools ensures all components operate within optimal parameters, supporting a robust and scalable solution for urban heat island analysis.

5.2 Infrastructure and Deployment

5.2.1 Overview

The Heat Island Analysis Tool is deployed on AWS infrastructure, utilizing various services to ensure scalability, reliability, and security. The system is containerized using Docker, orchestrated by Docker Compose, and managed using Ansible and Terraform for configuration and deployment. The overall infrastructure is depicted in Figure 5.6, showcasing the interplay between various components to achieve a seamless deployment process.

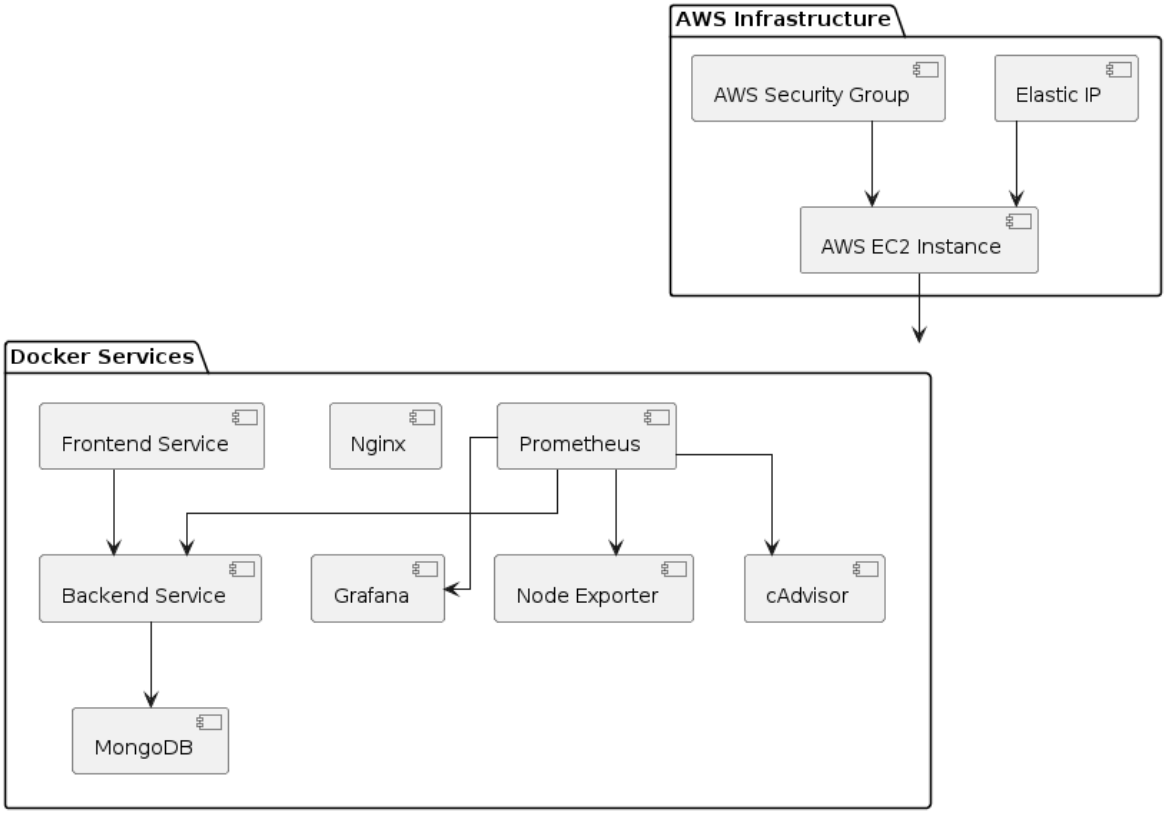


Figure 5.6: Overview of the Infrastructure

5.2.2 AWS EC2 Instance

The application is hosted on an AWS EC2 instance, which provides the computational resources required to run the application and its services. The EC2 instance is configured with an appropriate security group and an elastic IP for stable and secure access. The security group acts as a virtual firewall, controlling the inbound and outbound traffic to and from the instance. This ensures that only authorized traffic is allowed, enhancing the application's security. The elastic IP provides a static IP address, making it easier to access the application consistently.

The EC2 instance is provisioned using Terraform, which automates the process of creating and configuring AWS resources. This includes setting up the instance type, specifying the AMI, and associating the instance with the security group and elastic IP. By using Terraform, the infrastructure can be managed as code, allowing for repeatable and consistent deployments.

5.2.3 Docker

Docker is used to containerize the application, ensuring consistent runtime environments across different development and deployment stages. Docker containers

encapsulate the application code, dependencies, and configurations, making running the application on any system that supports Docker is easy. The Docker setup includes multiple services, such as the frontend, backend, and various monitoring tools, each running in its own container. The Docker services architecture, shown in Figure 5.7, highlights the interaction between these containers, ensuring isolated yet interconnected service operation.

Each service in the Docker architecture is defined in the `docker-compose.yml` file. This file specifies the configuration for each container, including the image to use, the ports to expose, and any environment variables required. Docker Compose is then used to manage the lifecycle of these containers, ensuring that they are started in the correct order and can communicate with each other. This approach simplifies the application services' deployment, scaling, and management.

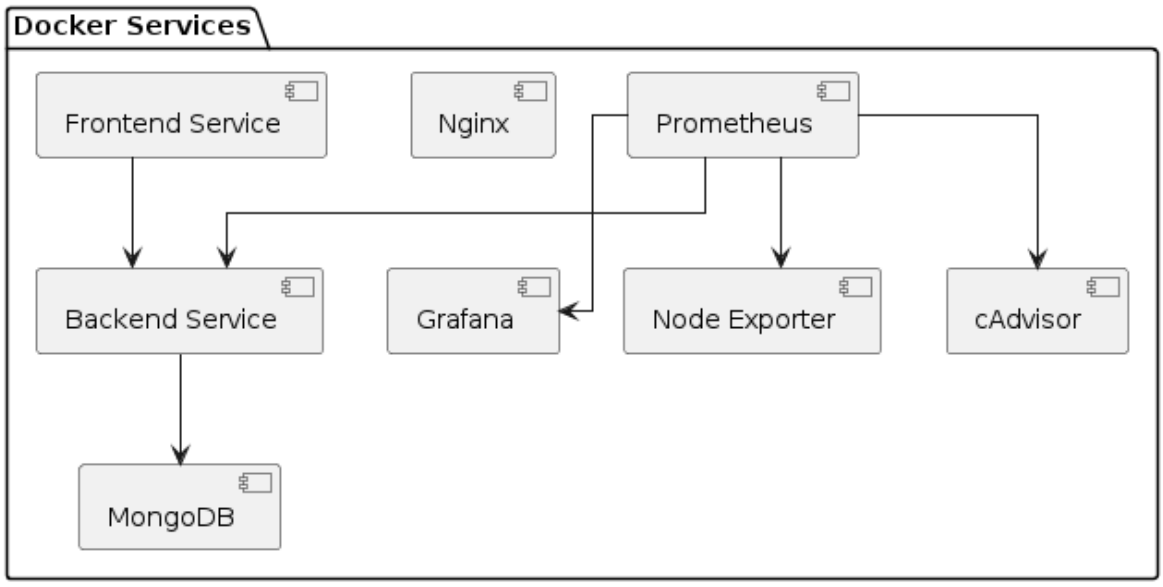


Figure 5.7: Docker Services Architecture

5.2.4 Security

Security is a critical aspect of the deployment, managed through AWS Security Groups. These security groups act as virtual firewalls for the EC2 instances, controlling the traffic that is allowed to reach the instances. Specific ports are opened for SSH, HTTP, and HTTPS access to ensure secure communication channels. The security configuration is illustrated in Figure 5.8, detailing the ports and protocols allowed through the security group.

The security group configuration is defined in the Terraform scripts. It includes rules to allow inbound traffic on port 22 for SSH access and ports 80 and 443 for

HTTP and HTTPS traffic, respectively. Additionally, all outbound traffic is allowed to ensure the instances can communicate with other services on the internet as needed. This setup ensures that the application is accessible while maintaining a secure environment. The use of security groups allows for fine-grained control over network traffic, enhancing the overall security posture of the deployment.

Furthermore, because the code is publicly available on a public repository in GitHub, it is essential to secure sensitive information such as keys and user-related data. To achieve this, `git-crypt` is used to encrypt these sensitive files. The `git-crypt` key is stored securely on a local machine, which is used to execute the Ansible scripts. During deployment, the Ansible playbook transfers this key to the EC2 instance, where it is used to decrypt the repository after it has been cloned. This ensures that sensitive information is not exposed publicly and is only accessible during the deployment process on the local machine.

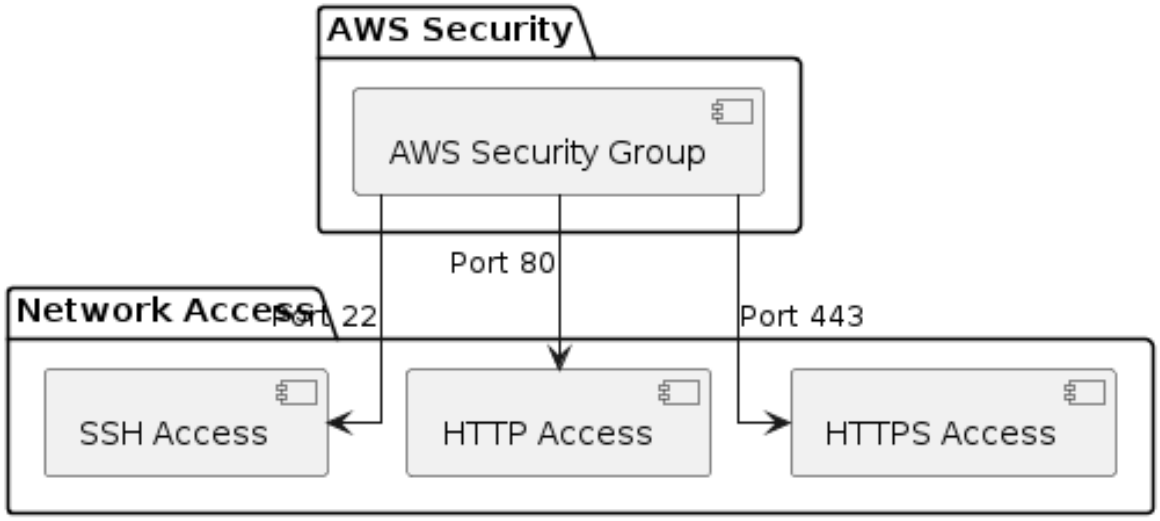


Figure 5.8: Security Configuration

5.2.5 Deployment

The deployment process leverages Ansible for configuration management and Terraform for infrastructure provisioning. Ansible scripts automate the setup and configuration of software on the EC2 instance, ensuring that all necessary packages and dependencies are installed correctly. Terraform scripts manage the creation and management of AWS infrastructure resources such as EC2 instances, security groups, and elastic IPs. The deployment architecture, depicted in Figure 5.9, shows how Ansible and Terraform work together to streamline the deployment process.

Terraform defines the infrastructure as code, allowing for the automated creation of AWS resources. Ansible configures the instances once the infrastructure is pro-

visioned, installing necessary software such as Docker, Docker Compose, and other dependencies. This ensures that the instances are ready to run the application containers as soon as they are started. By using Ansible and Terraform, the deployment process becomes automated, reducing the potential for human error and increasing the reliability of the deployment.

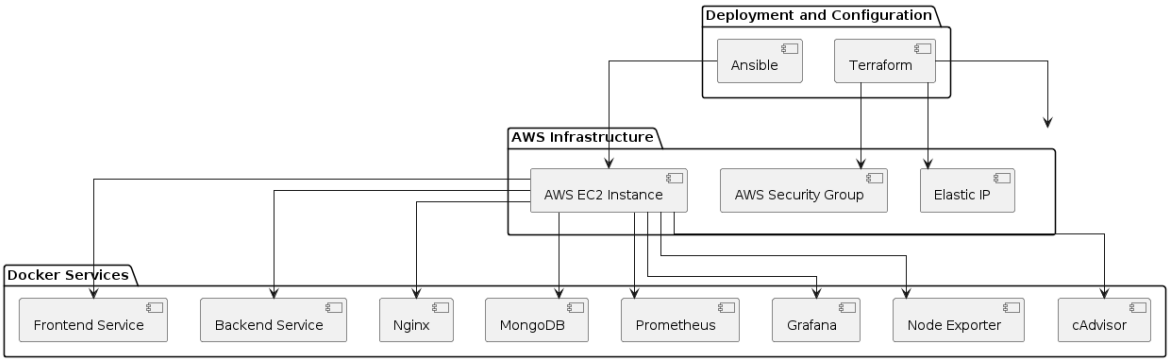


Figure 5.9: Deployment and Configuration Architecture

5.2.6 Orchestration

Docker Compose is used for orchestration, enabling the definition and running of multi-container Docker applications. Docker Compose ensures that all services are started in the correct order and are properly linked, facilitating communication between containers. This orchestration simplifies the management of the application's microservices architecture. The orchestration architecture is shown in Figure 5.10, illustrating how Docker Compose manages the lifecycle of the containers.

In the `docker-compose.yml` file, each service is defined with its configuration, including the Docker image, environment variables, ports, and dependencies. Docker Compose handles the orchestration, ensuring that dependent services are started in the correct order. For example, the backend service depends on the database service, so Docker Compose ensures that the database is started before the backend. This orchestration capability is crucial for maintaining the integrity and functionality of the application, as it manages dependencies and service interactions efficiently.

This comprehensive infrastructure and deployment strategy ensures that the Heat Island Analysis Tool is robust, scalable, and secure. The deployment process is streamlined by leveraging modern tools and methodologies, reducing complexity and increasing efficiency. This approach allows the development team to focus on enhancing the application's features and capabilities, confident that the underlying infrastructure is reliable and manageable.

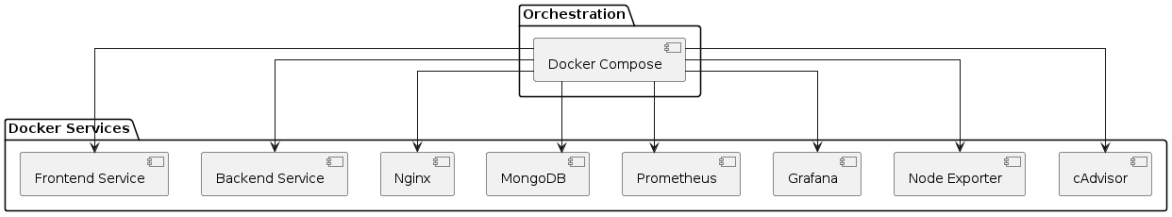


Figure 5.10: Orchestration with Docker Compose

5.3 Networking

5.3.1 Overview

The networking setup for the Heat Island Analysis Tool ensures secure and efficient communication between various application components. The use of an Elastic IP, Nginx for reverse proxy and load balancing, SSL certificates for secure communication, and security groups for controlling access are critical components. The overall networking architecture is depicted in Figure 5.11.

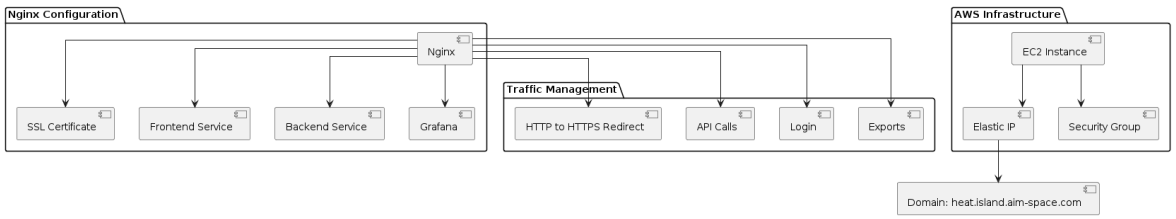


Figure 5.11: Networking Overview

5.3.2 Elastic IP

An Elastic IP is used to provide a static IP address for the EC2 instance hosting the application. This static IP ensures the application can be accessed reliably, even if the underlying instance is restarted or replaced. The Elastic IP is associated with the EC2 instance through Terraform scripts, which automate the provisioning and association process. This setup allows for consistent and reliable access to the application via the domain name heat.island.aim-space.com.

5.3.3 Nginx Configuration

Nginx is configured as a reverse proxy to manage incoming traffic and direct it to the appropriate backend or frontend services. The Nginx configuration file, `heat-island.conf`, contains directives for redirecting HTTP traffic to HTTPS, setting up SSL certificates,

and proxying requests to different services. The configuration ensures that all traffic is encrypted using TLS and that requests are efficiently routed to the correct service.

Key aspects of the Nginx configuration include:

- Redirecting all HTTP traffic to HTTPS to ensure secure communication.
- Using SSL certificates to encrypt traffic between clients and the server.
- Proxying requests to the frontend service at port 3000 and backend service at port 8080.
- Managing specific routes for API calls, login, and exports.
- Configuring a proxy for Grafana dashboards to monitor the application.

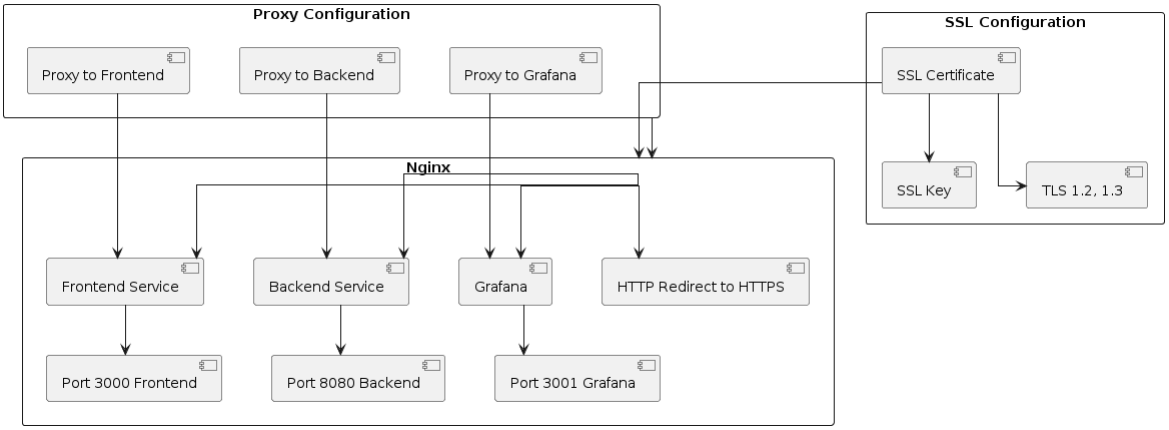


Figure 5.12: Nginx Configuration

5.3.4 SSL Certificate

SSL certificates are used to encrypt communication between clients and the server. The SSL certificate and key for the domain heat.island.aim-space.com are manually obtained from cPanel. The certificates are configured in the Nginx `heat-island.conf` file to ensure that all traffic is encrypted using TLS 1.2 and 1.3 protocols.

The Nginx configuration specifies the locations of the SSL certificate and key files, the supported SSL protocols, and the preferred ciphers for secure communication. By enforcing HTTPS, the application ensures that data transmitted between clients and the server remains confidential and secure.

Chapter 6

Conclusions

6.1 Importance of Integrated Analysis Tools

The development and implementation of the tool presented in this research highlight the increasing need for comprehensive urban heat island (UHI) analysis and mitigation strategies. As cities continue to grow, they face mounting challenges associated with UHIs, significantly impacting public health, energy consumption, and environmental sustainability. This tool, which integrates satellite data analysis with urban planning, offers a robust framework for addressing these challenges effectively.

6.1.1 Public Health and Environmental Impacts

Urban heat islands contribute to higher temperatures in cities compared to their rural surroundings, leading to increased heat-related illnesses and mortality rates, particularly among vulnerable populations such as the elderly and children. Elevated temperatures exacerbate air pollution levels, creating a feedback loop further degrading air quality and public health [10]. Addressing UHIs is crucial for improving the quality of life in urban areas and reducing health risks associated with extreme heat events.

The increased temperatures in urban areas result in higher demand for air conditioning, leading to greater energy consumption and higher costs for residents and businesses. This has economic implications and contributes to increased greenhouse gas emissions, further exacerbating climate change [17]. By providing detailed insights into the spatial distribution of heat islands, the tool can inform targeted interventions that enhance energy efficiency and reduce economic burdens.

6.2 Advancements Needed in Satellite Data

The research emphasizes the critical need for improved satellite data missions, focusing on enhanced resolution and real-time capabilities. Advanced satellite missions conducted by governmental agencies, non-governmental organizations (NGOs), or private companies are essential. Moreover, making some of this data freely available will significantly benefit research and practical applications in urban planning and environmental management [2].

6.2.1 Current Data Limitations

Current satellite resolutions present limitations that hinder detailed analysis. The existing satellite data, while valuable, often lacks the granularity needed for precise UHI detection and analysis. Higher resolution data and real-time capabilities would significantly enhance the accuracy and applicability of UHI detection and analysis tools. As discussed in Chapter 4, the limitations of current satellite technologies highlight the urgent need for advancements in this field 3.

6.2.2 Benefits of Enhanced Satellite Missions

Improved satellite data would provide more detailed and accurate information, enabling more precise identification and analysis of UHIs. Enhanced satellite missions with better spatial, spectral, and temporal resolution would support more effective mitigation strategies, contributing to better urban planning and environmental management. The availability of free satellite data would democratize access to critical information, fostering innovation and collaboration among researchers, policymakers, and urban planners [24, 33].

6.3 Advantages of Utilizing Google Earth Engine

The research demonstrates the significant benefits of using Google Earth Engine (GEE) for UHI analysis. GEE provides powerful tools for managing and processing large datasets efficiently, ensuring more accurate and reliable analysis of heat island effects.

6.3.1 Overcoming Cloud Interference

One of the major advantages of GEE is its ability to avoid cloud interference in data processing. The technical approach discussed in Chapter 5 illustrates how GEE's cloud masking functions ensure that only clear-sky pixels are used in the analysis,

significantly improving the accuracy of surface temperature readings 4. This capability is crucial for obtaining reliable data in regions with frequent cloud cover.

GEE's capability to handle cloud interference makes it a preferred choice for continuous monitoring and analysis of urban environments. By leveraging GEE's cloud computing power, researchers can process vast amounts of data more efficiently, leading to quicker and more informed decision-making processes.

6.3.2 Scalability and Integration Capabilities

GEE's scalability and ability to handle large datasets and complex computations make it ideal for analyzing extensive geographical areas over long periods. Additionally, GEE can integrate geospatial datasets, facilitating comprehensive environmental and climate studies. This integration capability allows for a holistic approach to UHI analysis, combining temperature data with land cover, vegetation indices, and other relevant datasets to provide a complete picture of urban heat dynamics [38].

The integration of diverse datasets not only enhances the accuracy of the analysis but also provides a multi-dimensional view of the urban heat phenomenon. This comprehensive approach is critical for developing effective mitigation strategies that address the root causes of UHIs.

6.4 Future Development

To further enhance the functionality and applicability of the tool, several areas for future development have been identified. These advancements promise to provide more comprehensive and actionable insights for urban planners and policymakers.

6.4.1 Air Quality Insights

Although air quality was mentioned in the state of the art review, it was not included in the current implementation. Future tool versions should integrate air quality data for a more comprehensive environmental analysis. By combining temperature and air quality data, the tool can better assess the impact of UHIs and air pollution on public health and the environment [7].

Including air quality data will enable the tool to offer more precise recommendations for improving urban air quality alongside reducing heat. This dual focus can significantly enhance public health outcomes and overall urban livability.

6.4.2 Cross-Analysis of Heat Islands

Expanding the analysis to include multiple heat islands across different areas or locations will provide deeper insights into the spatial dynamics and factors contributing to urban heat islands. Comparative studies between different cities or regions could reveal patterns and commonalities that are not apparent when analyzing a single location. This broader perspective will help identify best practices and effective mitigation strategies that can be applied in diverse urban contexts [16].

Cross-analysis of heat islands can uncover regional differences and similarities, allowing for tailored approaches to UHI mitigation. Understanding these nuances is essential for developing effective and context-specific strategies.

6.4.3 Development of a Specialized NLP Model

Developing a custom Natural Language Processing (NLP) model trained exclusively on mitigation solutions and real problems specific to various areas will enhance the tool's capability to recommend effective interventions. Such a model could analyze vast amounts of textual data, including scientific literature and policy documents, to identify the most effective strategies for specific contexts. This would enable the tool to provide tailored recommendations that address the unique challenges faced by different urban areas [36].

A specialized NLP model can streamline the process of identifying relevant mitigation strategies, making it easier for urban planners to access and implement the most appropriate solutions for their specific contexts.

6.4.4 Integration with Vehicle Traffic Data

Incorporating vehicle traffic data will allow a better understanding of its impact on heat islands, enabling more targeted mitigation strategies. Traffic patterns contribute significantly to urban heat, and understanding their role can help design more effective cooling measures. By integrating traffic data, the tool can identify hotspots of vehicular emissions and recommend interventions such as traffic management, public transportation improvements, and green infrastructure to mitigate their impact [21].

Understanding the interplay between traffic patterns and heat islands is crucial for developing comprehensive urban cooling strategies. Traffic management solutions can be integrated with other UHI mitigation measures to create a more holistic approach to urban heat reduction.

6.4.5 Analysis of Green Level

Analyzing the green levels in the areas studied will help assess the effectiveness of existing vegetation and identify opportunities for increasing green spaces to combat UHI effects. Urban greenery plays a crucial role in cooling urban areas, and detailed analysis can guide urban greening initiatives. By evaluating the distribution and health of urban vegetation, the tool can recommend strategic planting and maintenance practices that enhance the cooling benefits of green spaces [8].

Increasing urban green spaces not only helps to mitigate UHIs but also enhances biodiversity, improves air quality, and provides recreational areas for residents. The tool's analysis can support urban planners in making data-driven decisions to optimize green space development.

6.5 Concluding Remarks

The research conducted provides a foundational tool for urban heat island detection and mitigation, leveraging advanced satellite data and computational techniques. The necessity for enhanced satellite missions and the benefits of integrating various data sources underscore the potential for significant advancements in this field. The future developments outlined promise to expand the tool's capabilities, offering more comprehensive and actionable insights for urban planners and policymakers.

In summary, the tool represents a significant advancement in urban heat island analysis, providing a versatile and powerful resource for addressing the complex challenges posed by urbanization. This tool's continued evolution and enhancement will contribute to more sustainable and resilient urban environments.

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